



Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur

II Mid Term Examination, Dec.-2022

Semester:	III	Branch:	CS/DS/AI/IT
Subject:	AEM	Subject Code:	3CS/CDS/CAI/IT2-01
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q.1 A random variable X may assume 4 values with probabilities $\frac{1+3x}{4}$, $\frac{1-x}{4}$, $\frac{1+2x}{4}$, $\frac{1-4x}{4}$.

Find the condition on x so that these values represent the probability function of X.

- Q.2 If X is a random variable, show that $\text{Var}(X) = E(X^2) - E^2(X)$.

- Q.3 If X has uniform distribution in (-3, 3). Find $P(|X - 2| < 2)$

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q.4 The joint probability mass function of (X, Y) is given by $P(x, y) = k(2x+3y)$, $x = 0, 1, 2$; $y = 1, 2, 3$. Find

(i) Marginal probability distribution of X.

(ii) Marginal probability distribution of Y

- Q.5 If 10% of the pens manufactured by the company are defective, find the probability that a box of 12 pens contain

(i) Exactly two defective pens

(ii) At least two defective pens

- Q.6 Find Moment generating Function of Exponential Distribution

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q.7 Of a large group of men 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution find the mean height and standard deviation.

Given $\Phi(1.64) = 0.45$ and $\Phi(0.13) = 0.05$

- Q.8 Calculate the coefficient of correlation for the following data

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Solution of Question Paper

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Branch/Semester: ...C.S./AI/DS/IT-III	Subject: ...Adv. Engg. Mathematics	Subject Code: 3CS/CDS/IT/CAI-2-01
Duration: 1.5 hours	Date: 23/12/22	Session (I/II/III): ...I...
Submitted By: Dr. Vijay Kumar Singh	Max Marks: 20	

Ans 1. Total Probability = 1 ; let $P_1 = \frac{1+3x}{4}$; $P_2 = \frac{1-x}{4}$; $P_3 = \frac{1+2x}{4}$; $P_4 = \frac{1-4x}{4}$

$$\Rightarrow \frac{1+3x+1-x+1+2x+1-4x}{4} = 1$$

$\Rightarrow$ The given function is a probability function.

Now.

$$P_1 = \frac{1+3x}{4} \geq 0 \Rightarrow x \geq -\frac{1}{3} ; P_2 \geq 0 \Rightarrow x \leq 1$$

$$P_3 \geq 0 \Rightarrow x \geq -\frac{1}{2} ; P_4 \geq 0 \Rightarrow x \leq \frac{1}{4}$$

$\therefore$ The values of x for which a probability function is defined lie in the range $-\frac{1}{3} \leq x \leq \frac{1}{4}$. Ans

Ans. 2. $\text{Var}(X) = E[(X-\mu)^2] ; \mu = E[X]$

$$= E[X^2 + \mu^2 - 2X\mu]$$

$$= E[X^2] + \mu^2 - 2\mu E[X]$$

$$= E[X^2] - \mu^2$$

$$= E[X^2] - (E[X])^2$$

Ans. 3. For Uniform distribution pdf $f(x) = \frac{1}{3+3} = \frac{1}{6}$

$$P[|X-2| < 2] = P[0 < X < 4] = \int_0^4 \frac{1}{6} dx = \frac{1}{2} \quad \underline{\text{Ans}}$$



Solution of Question Paper

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Branch/Semester: <u>SS/IT/AI/DS - III</u>	Subject: <u>.....AEM</u>	Subject Code: <u>3.CS/IT/CDS/CAI2-01</u>
Duration: 1.5 hours	Date: <u>28/12/22</u> Session (I/II/III): <u>I.....</u>	Max Marks: <u>20</u>
Submitted By: <u>Dr. V. K. Singhal</u>		

Ans 4

$X \backslash Y \rightarrow$	1	2	3
$\downarrow$			
0	3K	6K	9K
1	5K	8K	11K
2	7K	10K	13K

$$\therefore \text{Total Prob} = 1 \Rightarrow K = \frac{1}{72}$$

(i) Marginal distribution of X

$X \downarrow$	Marg. Distrib. of X
0	$18/72$
1	$24/72$
2	$30/72$

(ii) Marginal distribution of Y

$Y \downarrow$	Marg. distrib. of Y
1	$15/72$
2	$24/72$
3	$33/72$

Ans 5. Given $p = 0.1 \Rightarrow q = 0.9$

Here $n = 12$



Solution of Question Paper

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Branch/Semester: <u>CS/IT/DS/AI - III</u>	Subject: <u>AEM</u>	Subject Code: <u>2CS/IT/DS/CA22-01</u>
Duration: 1.5 hours	Date: <u>23/12/22</u> Session (I/II/III): <u>I</u>	Max Marks: <u>20</u>
Submitted By: <u>Dr. V.K. Singhal</u>		

$$(i) P(X=2) = {}^{12}C_2 (0.1)^2 (0.9)^{10} = 0.2301$$

$$\begin{aligned} (ii) P(X \geq 2) &= 1 - P(X=0) - P(X=1) \\ &= 1 - {}^{12}C_0 (0.1)^0 (0.9)^{12} - {}^{12}C_1 (0.1)^1 (0.9)^{11} \\ &= 1 - 0.2824 - 0.3766 \\ &= 0.341 \end{aligned}$$

Ans. 6. We know that Pdf of Exponential distribution

$$f(x) = \lambda e^{-\lambda x}; x > 0$$

$$\begin{aligned} \text{Mgf of } X &= M_X(t) = E[e^{tx}] \\ &= \int_0^{\infty} e^{tx} (\lambda e^{-\lambda x}) dx \\ &= \lambda \int_0^{\infty} e^{-x(\lambda-t)} dx \\ &= \lambda \left[\frac{e^{-x(\lambda-t)}}{-(\lambda-t)} \right]_0^{\infty} \end{aligned}$$

Solution of Question Paper

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Duration: 1.5 hours	Date: <u>23/12/22</u> Session (I/II/III): <u>I.....</u>	Max Marks: <u>20</u>
Submitted By: <u>Dr. V. K. Singhal</u>		

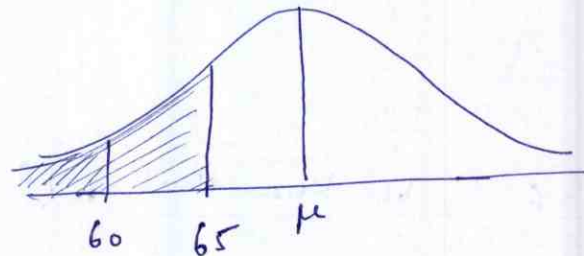
$$\begin{aligned}
 &= \frac{1}{1-t} \\
 &= \left(1 - \frac{t}{1}\right)^{-1} \\
 &= 1 + \frac{t}{1} + \frac{t^2}{1^2} + \dots
 \end{aligned}$$

Ans 7. Given $P(X < 60) = 0.05$

~~$P(X < 60)$~~ $P(60 < X < 65) = 0.40$

Let Mean = μ & SD = σ

$\therefore X = 60$ and 65 both lies to the left of mean μ . let $Z = \frac{X - \mu}{\sigma}$



let at $X = 60$ $Z = -Z_1 \Rightarrow -Z_1 = \frac{60 - \mu}{\sigma}$ — (1)

at $X = 65$ $Z = -Z_2 \Rightarrow -Z_2 = \frac{65 - \mu}{\sigma}$ — (2)

$\therefore$ Given $P(X < 60) = P(Z < -Z_1) = 0.05$

$\Rightarrow P(Z > Z_1) = 0.05$ (Using symmetry of normal curve)

$\Rightarrow 0.5 - P(0 < Z < Z_1) = 0.05$

$\Rightarrow \phi(Z_1) = 0.45$

$\Rightarrow Z_1 = 1.64$

$\therefore$ by eq (1) $-1.64 = \frac{60 - \mu}{\sigma}$ — (3)



Solution of Question Paper

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Duration: 1.5 hours	Date: <u>23/12/22</u> Session (I/II/III) <u>I</u>	Max Marks: <u>20</u>
Submitted By: <u>Dr. V.K. Singhal</u>		

Also given $P(60 < X < 65) = 0.40$

$$\Rightarrow P(-z_1 < Z < -z_2) = 0.40$$

$$\Rightarrow P(z_2 < Z < z_1) = 0.40$$

$$\Rightarrow \phi(z_1) - \phi(z_2) = 0.40$$

$$\Rightarrow \phi(z_2) = \phi(z_1) - 0.40$$

$$= 0.45 - 0.40$$

$$= 0.05$$

$$\therefore z_2 = 0.13$$

$\therefore$ by eqⁿ ③

$$-0.13 = \frac{65 - \mu}{\sigma}$$

— (4)

solving eqⁿ ③ & ④

$$\mu = 65.43$$

$$\text{and } \sigma = 3.30$$

Ans

Ans. 8. we know that correlation coefficient is

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{where } \text{Cov}(X, Y) = \frac{\sum xy}{n} - \left(\frac{\sum x}{n}\right) \left(\frac{\sum y}{n}\right)$$

$$\sigma_X = \sqrt{\left(\frac{\sum x^2}{n}\right) - \left(\frac{\sum x}{n}\right)^2} ; \sigma_Y = \sqrt{\left(\frac{\sum y^2}{n}\right) - \left(\frac{\sum y}{n}\right)^2}$$

Solution of Question Paper

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Branch/Semester: <u>CS/IT/DS/AI - III</u>	Subject: <u>AEM</u>	Subject Code: <u>355/IT/CAI/CD-2-01</u>
Duration: 1.5 hours	Date: <u>23/12/22</u> Session (I/II/III): <u>I</u>	Max Marks: <u>20</u>
Submitted By: <u>D. V. K. Singhal</u>		

Here $\bar{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$

$\bar{y} = \frac{108}{9} = 12$

x	y	$u = x - 5$	$v = y - 12$	u^2	v^2	uv
1	9	-4	-3	16	9	12
2	8	-3	-4	9	16	12
3	10	-2	-2	4	4	4
4	12	-1	0	1	0	0
5	11	0	-1	0	1	0
6	13	1	1	1	1	1
7	14	2	2	4	4	4
8	16	3	4	9	16	12
9	15	4	3	16	9	12
Total		0	0	60	60	57

We know that $r(x, y) = r(u, v) = \frac{\text{Cov}(u, v)}{\sigma_u \sigma_v}$

Now $\text{Cov}(u, v) = \frac{\sum uv}{n} - \frac{\sum u}{n} \cdot \frac{\sum v}{n}$
 $= \frac{57}{9} - 0 = 6.333$



Solution of Question Paper

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Submitted By: Dr. V. K. Singhal		

$$\sigma_u = \sqrt{\frac{\sum u^2}{n} - (\bar{u})^2} = \sqrt{6.666} = 2.582$$

$$\sigma_v = 2.582$$

Hence

$$r = \frac{\text{Cov}(U, V)}{\sigma_u \sigma_v} = \frac{6.333}{(2.582)^2} = 0.95$$

Ans



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II Mid Term Examination, Dec.-2022**

Semester:	III	Branch:	CS/AI/DS
Subject:	Technical Communication	Subject Code:	3CS/AI/DS 1-02
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): II		Date: 22/12/22	

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. Mention few characteristic features of a technical report.
Q2. Differentiate between Abstract and Summary.
Q3. Mention some types of Technical Discourses.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Mention and describe the components of a Technical Report.
Q5. Define Minutes of the meeting and its significance.
Q6. Mention some points which should be kept in mind for designing a document effectively.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q7. Discuss different steps involved in the writing of a technical text.
Q-8. Write down the format or structure of a Memorandum?



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Duration: 1.5 hours	Date :22/12/22 Session : (II)	Max Marks:20
Submitted By: Dr Shipra Malik		

PART-A (Attempt all the questions.)

3x2=6 marks

Ans.1

i. Clarity

Technical document must convey a single meaning that the reader can understand.

ii. Accuracy

Unclear writing can cause many problems and even inaccuracy in the report. If you mean to write 40,000 don't write 400,000. If the reader suspects that you are slanting information they have the right to doubt the entire document.

3. Comprehensiveness:

When writing technically, all the information should be provided, its background must be described and clear description of any process, or method of carrying out a specific work, should also be given. It also includes results, conclusions and recommendations.

4. Accessibility:

It means the ease with which the readers can locate the information they seek. To increase Accessibility, include headings and lists in the report. A table of contents, list of illustrations glossary and index are preferred.

5. Conciseness:

Technical writing is meant to be useful. The longer a document is, the more difficult it gets to use it. Even it takes more of the user's time. Conciseness works against clarity and comprehensiveness.

Ans2. While an abstract is a short, descriptive paragraph overviewing your entire paper from introduction to the findings or future studies, a summary includes your entire paper and its visuals, just in a shorter length and more concise than its original document.

Ans 3. i.Descriptive Discourse

A descriptive discourse often takes two forms; it can be in static form, or the form called process description.

The static description draws a verbal picture using words that appeal to the senses, while the process



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description tends to explain the various degrees or levels of advancement involved in carrying out a task.

ii. Narrative Discourse

A narrative discourse is that which in its description, portrays causally related incidents; here the occurred incidents are often arranged one after another in an order of chronology.

iii. Expository Discourse

An expository discourse consists in giving definitive explanation and clarification by means of examples and illustrations, details, definition, and other rhetorical devices of like nature.

iv. Argumentative Discourse

The argumentative discourse is used with the sole purpose of persuading the audience (hearers or readers) to either accept or reject opinions. As a primary prerequisite, argumentative discourse only takes effect where there is a contentious or controversial topic.

PART-B (Attempt any two)

2x4=8 marks

Title page

Technical reports usually require a title page. To know what to include, follow the conventions required in your subject.

Table of contents

This is optional depending on the length of the report—a 2 to 3 page report probably wouldn't have a table of contents but a 10 to 20 page report would.

Introduction

A technical report introduction:

- provides context for the problem being addressed,
- discusses relevant previous research, and
- states your aim or hypothesis.

Body



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The body of a technical report is structured according to the needs of your reader and the nature of the project. The writer decides how to structure it and what to include.

To help, ask yourself:

- What does the reader need to know first?
- What is the most logical way to develop the story of the project?

Figures, tables, equations and formulae /List of Illustrations:

Technical reports include a mixture of text, tables, figures and formulae. Consider how you can present the information best for your reader. Would a table or figure help to convey your ideas more effectively than a paragraph describing the same data?

Figures and tables should:

- Be numbered
- Be referred to in-text, e.g. *In Table 1...*, and
- Include a simple descriptive label - above a table and below a figure.

Conclusion

Your conclusion should mirror your introduction.

Be sure to:

- Refer to your aims
- Summarise your key findings, and
- State your major outcomes and highlight their significance.

Recommendations

- If your technical report includes recommendations for action. You could choose to report these as a bullet point list. When giving an answer to your problem, be sure to include any limitations to your findings.



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References

- Acknowledge all the information and ideas you've incorporated from other sources into your paper using a consistent referencing style.

Appendices

If you have data that is too detailed or lengthy to include in the report itself, include it in the appendix. Your reader can then choose to refer to it if they are interested. Label your appendix with a number or a letter, a title, and refer to it the text

Ans 5. Meeting minutes are **notes that are recorded during a meeting**. They highlight the key issues that are discussed, motions proposed or voted on, and activities to be undertaken. The minutes of a meeting are usually taken by a designated member of the group. Some points regarding its importance are:

1. Minutes help save time and money

It might appear that writing meeting minutes can take a lot of time, they will, actually, assist you to save time and money. Meeting minutes will give you a written record of what was examined and acknowledged at a meeting, so you and your partners will have similar recollections from the meeting and the same thoughts regarding what was agreed.

2. Minutes offer legal protection and is a proof as it is a written documentation or record

Meeting minutes are an official record of corporate business. They give a record of activity or proof of interaction.

PART-C (Attempt any one)

1x6=6 marks

Ans 6. Titles

Almost every document that exists as a standalone unit must have a title that accurately represents its contents in a nutshell. It's the first thing a reader looks for to understand what a document is all about and should thus be easily found centred at the top of the first page of any small document, and prominently placed on the cover



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Duration: 1.5 hours	Date :22/12/22 Session : (II)	Max Marks:20
Submitted By: Dr Shipra Malik		

of larger documents.

- **Topic summary:** A title is the most concise summary possible of a topic while still making sense. If you glance at a news website or newspaper, for instance, you can get a reasonably good sense of what's going on in the world just by reading the headlines because they are titles that, in as few words as possible, summarize the narratives told in the articles that follow.
- **Conciseness:** Aim for a length in the 2- to 7-word range—something that can be said repeatedly in one short breath. One-word titles are appropriate only for art (e.g., for books, films, songs, albums, etc.), but most other professional documents use a reasonable number of words to give a sense of the topic, albeit streamlined to the point of having no words that don't absolutely need to be there. In scientific papers, titles can be quite long and carry plenty of detail, though you can expect that their audiences will rarely pronounce the full title.
- **Capitalization:** Capitalize the first word no matter what, as well as all major words (nouns, verbs, adjectives, adverbs, pronouns, etc.) thereafter.
 - Don't capitalize prepositions (e.g., *on, to, from, in, out, of*), conjunctions (*and, but, or, for, nor, so, yet*), nor articles (*the, a, an*) unless they're the first word of the main title or subtitle (Darling, 2014a, 2014b, 2014c).
 - When including a hyphenated word (i.e., with a compound-modifier hyphen), leave the second word, the one immediately following the hyphen, lowercase (see the first two examples in the titles listed at the end of this subsection).
- **Structure:** Use a noun, verb, or adjective phrase rather than a complete sentence.
 - **Main title:** If your title comes in two parts with a main title and subtitle, the main title establishes the general context of the topic, perhaps with catchy or clever phrasing, and ends with a colon (:) with a single space after it but none before.
 - **Subtitle:** The subtitle follows the main title with a more specific and detailed summary of the document topic.
- **Position:** Centre the title at the top of the page and include 1-2 empty lines below it to separate it from the opening text.
- **Typeface:** Use bold typeface to help draw the eye towards the title, as well as colour if appropriate.

Ans 7. Collecting information — the first step in the writing process

This step consists in collecting every piece of information you can find about the product you are about to describe. This is the most important part of the process because all the subsequent steps and the whole content rely on this preliminary phase.

During this phase, you must carry out an in-depth analysis of the content and the context. For instance : scope, schedule, the legal framework of the target market, audience, and last but not least, the product and how to use it.



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2. Writing needs structure

During the structuring phase, you will create a well-defined Table of Contents. You need to dedicate time to establish a clear structure because this will be the backbone of your writing journey. However, you should be prepared to modify it along the way, as some fine-tuning may be needed.

3. Drafting —put into practice your technical writing skills

This is the step where you really get into action. As a technical writer, you must always consider the user, and adopt an action-oriented — or “task-oriented” — approach, bearing in mind that your reader is a user who is looking to accomplish a task. After the organization step, you’re ready for a first draft. The draft will be a base for your future documentation, create it as it’s comfortable for you, for example, you may use long descriptions, not formatted lists and so on, but then you should revise the written content wisely — follow all the technical writing rules like placing important information first, using the clear sentence structure, active voice and so on.

Review and Publish

The last step is reviewing and publishing your content. You can use these proofreading tips for technical writers to polish your documentation, use free grammar checkers or/and ask your friend or colleague to read and follow some steps of your documentation. After that, you’re ready to send your document to reviewers. When it’s ok, publish your documentation. But it’s not like publish and forget about that. You need to examine analytics constantly to improve your content. There are different metrics that you can use to see whether everything is going according to your plan.

Ans8. A memo is usually composed of a **heading, an introduction, a discussion, and a conclusion**. Supporting documents or visuals are sometimes attached to a memo.

Generally, an internal memo format includes these sections:

TO: names of recipients

FROM: name of the sender

DATE / TIME: to track messages

RE: or SUBJECT: purpose of the memo

MESSAGE or BODY: contains the relevant information of the memo.

SIGNATURE: The originator of the message signs the memo.



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Ramnagar, Jagatpura, Jaipur-302017

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CC: recipients who need to be apprised of the contents of a message.

ATTACHMENTS: indicates other documents have been added to the memo.



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Semester:	III	Branch:	CSE, AI, DS
Subject:	DE	Subject Code:	3CAI3-04, 3CS3-04, 3DS3-04
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q.1 Solve the following:

(a) Determine the output frequency for a frequency division circuit that contains 10 flip-flops with an input clock frequency of 24.48 Mhz.

(b) Draw the circuit diagram of J-K flip flop and write its truth table.

(c) Design a 3 to 8 decoder using 2 to 4 decoders.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q.2 Implement the following function using 4x 1 multiplexer.

$$f(A, B, C) = \sum m(0, 2, 5, 6)$$

Q.3 Describe the characteristics of Logic Families.

Q.4 Design a 4 bit universal shift register and explain its working in all the modes of operations.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q.5 Design a 4bit look ahead carry adder.

Q.6 Using JK flip flop design the synchronous counter that counts in the sequence: 001, 100, 010, 101, 110, 111, 011, 001...



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Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 21.12.2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shanu Tripathi, Dr. Vinay Kanungo, Dr. Niketa Sharma, Vinod Kataria		

Ans. 1

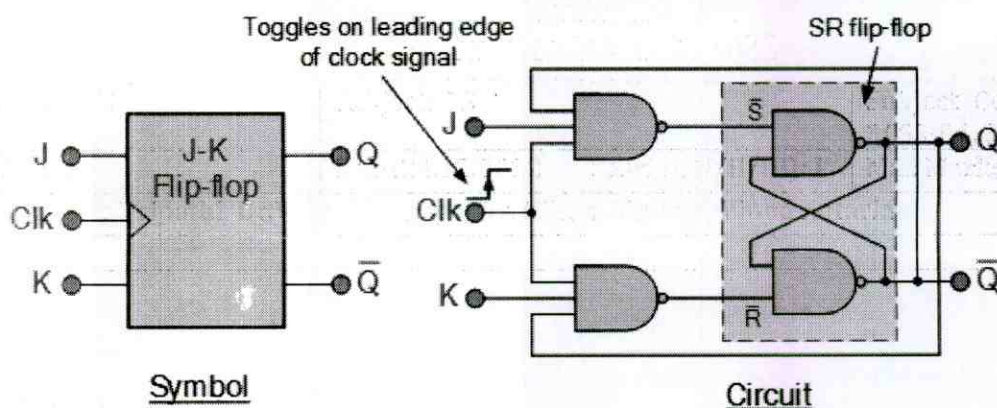
(a)

No of flip flops = 10

Total no of states = $2^{10} = 1024$

Output clock frequency = $24.48 \times 10^6 / 1024 = 23.9 \text{ khz}$

(b)



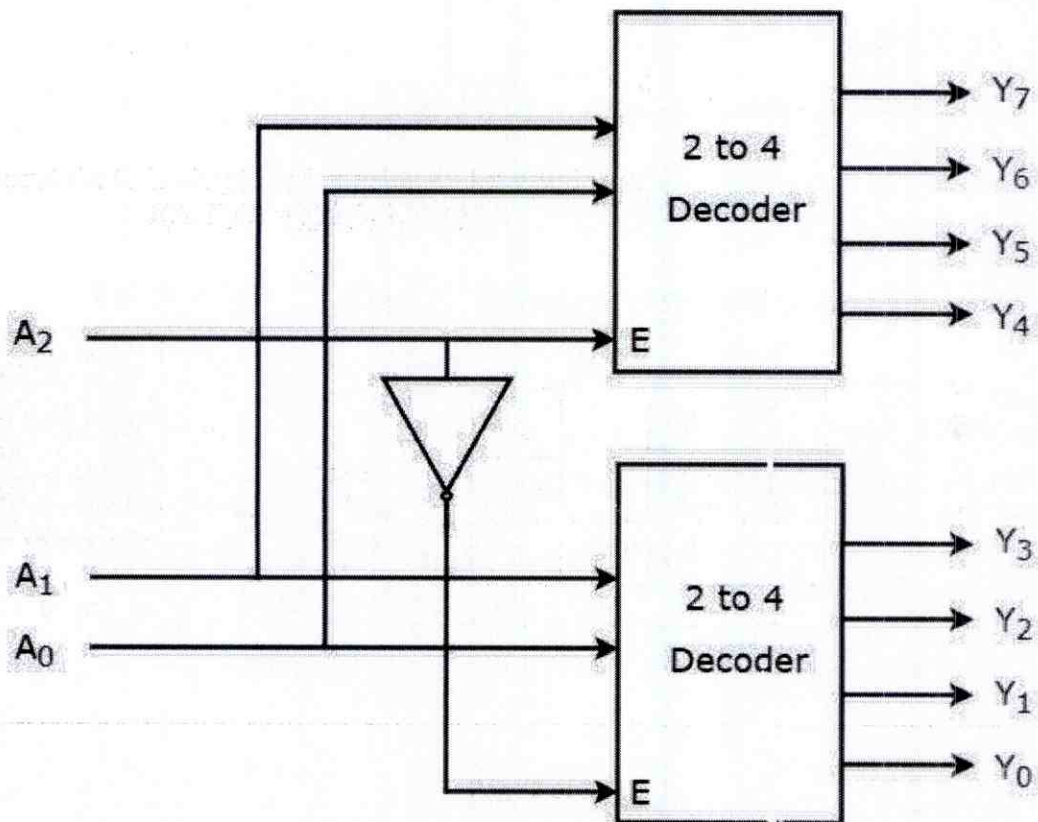
Same as for SR Latch	Clock	Input		Output		Description
	Clk	J	K	Q	Q'	
	X	0	0	1	0	Memory no change
	X	0	0	0	1	
	$\downarrow$	0	1	1	0	Reset Q>>0
	X	0	1	0	1	
	$\downarrow$	1	0	0	1	Set Q>>1
	X	1	0	1	0	
Toggle action	$\downarrow$	1	1	0	1	Toggle
	$\downarrow$	1	1	1	0	



Solution of Question Paper
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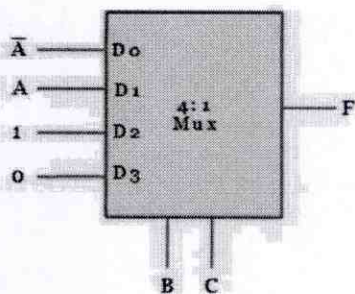
Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date:21.12.2022 Session (I/II/III): I	Max Marks:20
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(c)



Q.2 Implement the following function using 4x 1 multiplexer.
 $f(A, B, C) = \sum m(0, 2, 5, 6)$

Ans. 2



	D ₀	D ₁	D ₂	D ₃
Row 1 $\bar{A}$	0	1	2	3
Row 2 A	4	5	6	7
	$\bar{A}$	A	1	0



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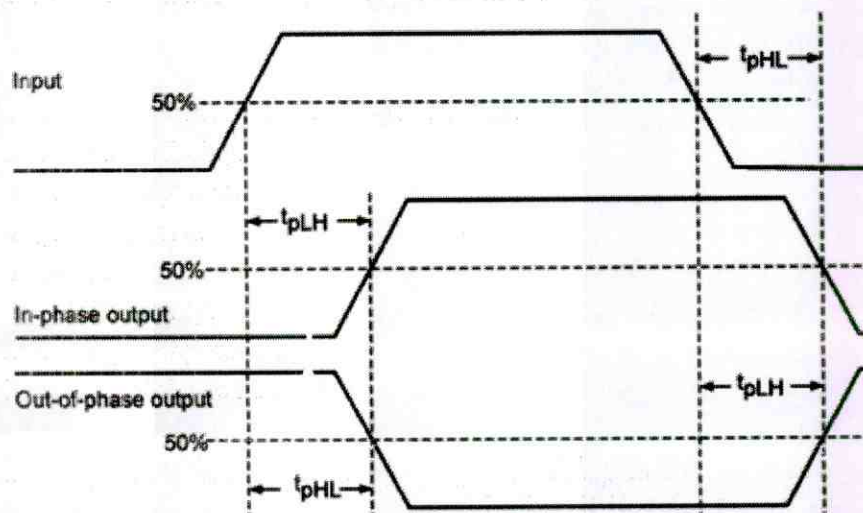
Q.3 Describe the characteristics of Logic Families.

Ans 3.

Following are the characteristics of logic families:

- 1) Propagation Delay.
- 2) Power Dissipation.
- 3) Figure of Merit.
- 4) Fan-Out.
- 5) Noise Immunity.
- 6) Operating Temperature.
- 7) Current & Voltage Parameters

Propagation delay t_p . The propagation delay is the time delay between the occurrence of change in the logical level at the input and before it is reflected at the output. It is the time delay between the specified voltage points on the input and output waveforms. Propagation delays are separately defined for LOW-to-HIGH and HIGH-to-LOW transitions at the output. In addition, we also define enable and disable time delays that occur during transition between the high-impedance state defined logic LOW or HIGH states.





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Power dissipation. The power dissipation parameter for a logic family is specified in terms of power consumption per gate and is the product of supply voltage V_{CC} and supply current I_{CC} . The supply current is taken as the average of the HIGH-level supply current I_{CCH} and the LOW-level supply current I_{CCL} .

Speed-power product. The speed of a logic circuit can be increased, that is, the propagation delay can be reduced, at the expense of power dissipation. We will recall that, when a bipolar transistor switches between cut-off and saturation, it dissipates the least power but has a large associated switching time delay. On the other hand, when the transistor is operated in the active region, power dissipation goes up while the switching time decreases drastically. It is always desirable to have in a logic family low values for both propagation delay and power dissipation parameters. A useful figure-of-merit used to evaluate different logic families is the speed-power product, expressed in picojoules, which is the product of the propagation delay (measured in nanoseconds) and the power dissipation per gate (measured in milliwatts).

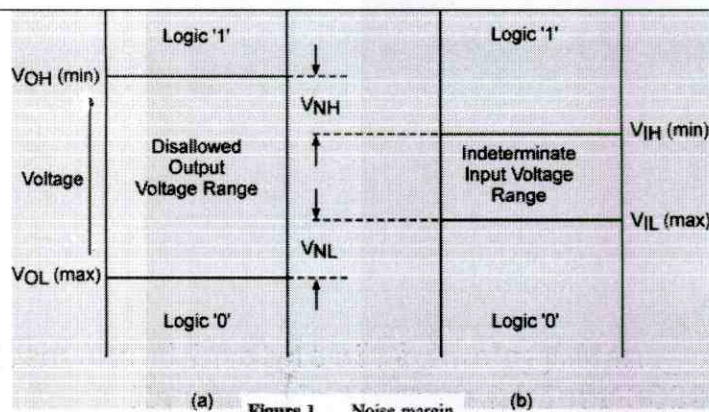
Fan-out. The fan-out is the number of inputs of a logic function that can be driven from a single output without causing any false output. It is a characteristic of the logic family to which the device belongs. It can be computed from I_{OH}/I_{IH} in the logic HIGH state and from I_{OL}/I_{IL} in the logic LOW state. If, in a certain case, the two values I_{OH}/I_{IH} and I_{OL}/I_{IL} are different, the fan-out is taken as the smaller of the two. This description of the fan-out is true for bipolar logic families like TTL and ECL. When determining the fan-out of CMOS logic devices, we should also take into consideration how much input load capacitance can be driven from the output without exceeding the acceptable value of propagation delay.

Noise margin. This is a quantitative measure of noise immunity offered by the logic family. When the output of a logic device feeds the input of another device of the same family, a legal HIGH logic state at the output of the feeding device should be treated as a legal HIGH logic state by the input of the device being fed. Similarly, a legal LOW logic state of the feeding device should be treated as a legal LOW logic state by the device being fed. We have seen in earlier paragraphs while defining important characteristic parameters that legal HIGH and LOW voltage levels for a given logic family are different for outputs and inputs. Figure 1 shows the generalized case of legal HIGH and LOW voltage levels for output [Fig. 1 (a)] and input [Fig. 1 (b)]. As we can see from the two diagrams, there is a disallowed range of output voltage levels from $V_{OL}(\text{max.})$ to $V_{OH}(\text{min.})$ and an indeterminate range of input voltage levels from $V_{IL}(\text{max.})$ to $V_{IH}(\text{min.})$. Since $V_{IL}(\text{max.})$ is greater than $V_{OL}(\text{max.})$, the LOW output state can therefore tolerate a positive voltage spike equal to $V_{IL}(\text{max.}) - V_{OL}(\text{max.})$ and still be a legal LOW input. Similarly, $V_{OH}(\text{min.})$ is greater than $V_{IH}(\text{min.})$, and the HIGH output state can tolerate a negative voltage spike equal to $V_{OH}(\text{min.}) - V_{IH}(\text{min.})$ and still be a legal HIGH input. Here, $V_{IL}(\text{max.}) - V_{OL}(\text{max.})$ and $V_{OH}(\text{min.}) - V_{IH}(\text{min.})$ are respectively known as the LOW-level and HIGH-level noise margin.



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Q.4 Design a 4 bit universal shift register and explain its working in all the modes of operations.

Ans. 4

Universal Shift Register: A Universal shift register is a register which has both the right shift and left shift with parallel load capabilities. Universal shift registers are used as memory elements in computers. A Unidirectional shift register is capable of shifting in only one direction. A bidirectional shift register is capable of shifting in both the directions. The Universal shift register is a combination design of **bidirectional** shift register and a **unidirectional** shift register with parallel load provision.

Basic connections:

- The first input (zeroth pin of multiplexer) is connected to the output pin of the corresponding flip-flop.
- The second input (first pin of multiplexer) is connected to the output of the very-previous flip flop which facilitates the right shift.
- The third input (second pin of multiplexer) is connected to the output of the very-next flip-flop which facilitates the left shift.
- The fourth input (third pin of multiplexer) is connected to the individual bits of the input data which facilitates parallel loading.
- The working of the Universal shift register depends on the inputs given to the select lines.

The register operations performed for the various inputs of select lines are as follows:

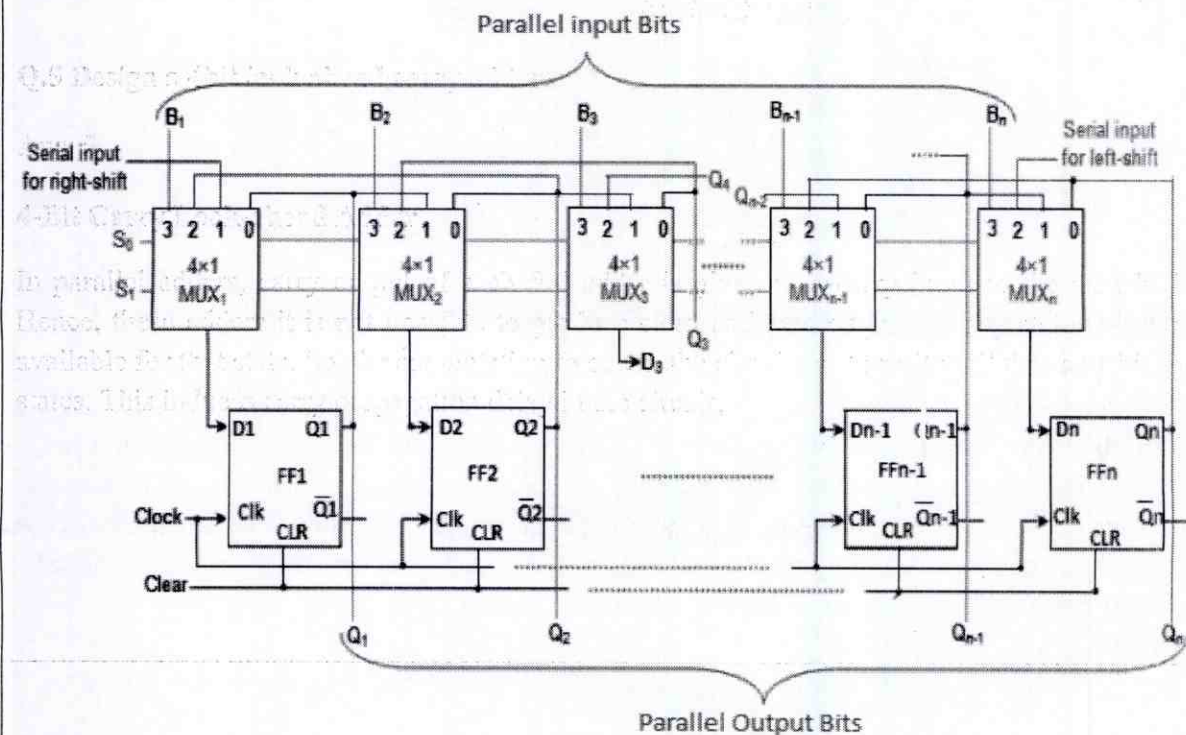
S1	S0	Mode of Operation
0	0	No Change
0	1	Right shift
1	0	Left Shift
1	1	Parallel Load



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Q.5 Design a 4bit look ahead carry adder.

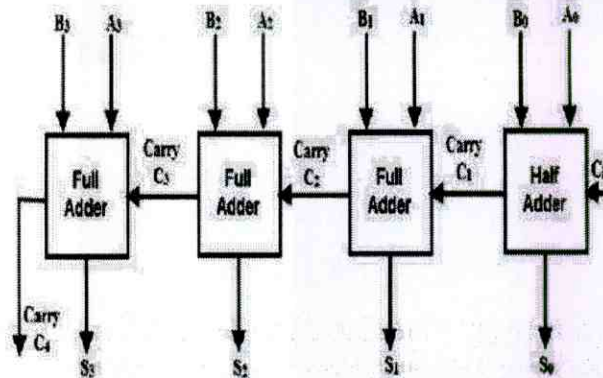
Ans. 5

4-Bit Carry Look-ahead Adder

In parallel adders, carry output of each full adder is given as a carry input to the next higher-order state. Hence, these adders it is not possible to produce carry and sum outputs of any state unless a carry input is available for that state. So, for computation to occur, the circuit has to wait until the carry bit propagated to all states. This induces carry propagation delay in the circuit.

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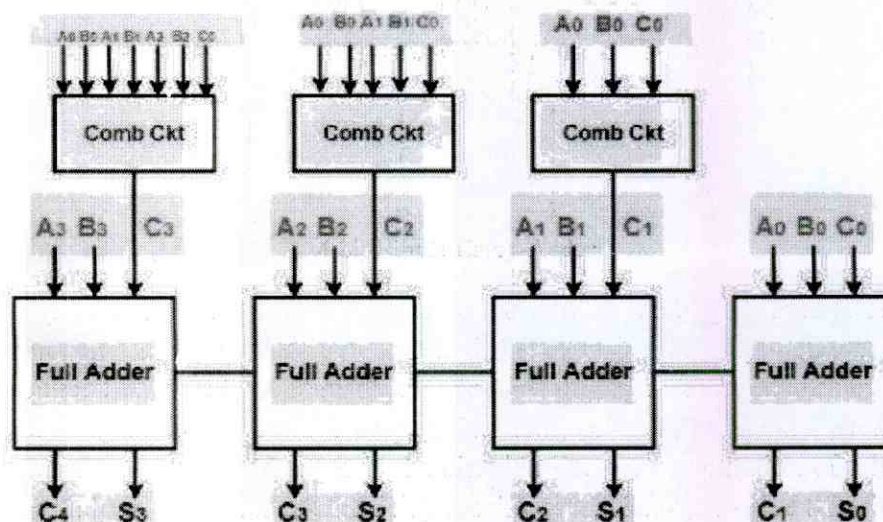
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4-bit-Ripple-Carry-Adder

Consider the 4-bit ripple carry adder circuit above. Here the sum S₃ can be produced as soon as the inputs A₃ and B₃ are given. But carry C₃ cannot be computed until the carry bit C₂ is applied whereas C₂ depends on C₁. Therefore to produce final steady-state results, carry must propagate through all the states. This increases the carry propagation delay of the circuit. The propagation delay of the adder is calculated as “the propagation delay of each gate times the number of stages in the circuit”. For the computation of a large number of bits, more stages have to be added, which makes the delay much worse. Hence, to solve this situation, Carry Look-ahead Adder was introduced.

To understand the functioning of a Carry Look-ahead Adder, a 4-bit Carry Look-ahead Adder is described below.



4-bit-Carry-Look-ahead-Adder-Logic-Diagram



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In this adder, the carry input at any stage of the adder is independent of the carry bits generated at the independent stages. Here the output of any stage is dependent only on the bits which are added in the previous stages and the carry input provided at the beginning stage. Hence, the circuit at any stage does not have to wait for the generation of carry-bit from the previous stage and carry bit can be evaluated at any instant of time.

Truth Table of Carry Look-ahead Adder

For deriving the truth table of this adder, two new terms are introduced – Carry generate and carry propagate. Carry generate $G_i = 1$ whenever there is a carry C_{i+1} generated. It depends on A_i and B_i inputs. G_i is 1 when both A_i and B_i are 1. Hence, G_i is calculated as $G_i = A_i \cdot B_i$.

Carry propagated P_i is associated with the propagation of carry from C_i to C_{i+1} . It is calculated as $P_i = A_i \oplus B_i$. The truth table of this adder can be derived from modifying the truth table of a full adder.

Using the G_i and P_i terms the Sum S_i and Carry C_{i+1} are given as below –

- $S_i = P_i \oplus G_i$.
 - $C_{i+1} = C_i \cdot P_i + G_i$.
- Therefore, the carry bits C_1, C_2, C_3 , and C_4 can be calculated as
- $C_1 = C_0 \cdot P_0 + G_0$.
 - $C_2 = C_1 \cdot P_1 + G_1 = (C_0 \cdot P_0 + G_0) \cdot P_1 + G_1$.
 - $C_3 = C_2 \cdot P_2 + G_2 = (C_1 \cdot P_1 + G_1) \cdot P_2 + G_2$.
 - $C_4 = C_3 \cdot P_3 + G_3 = C_0 \cdot P_0 \cdot P_1 \cdot P_2 \cdot P_3 + P_3 \cdot P_2 \cdot P_1 \cdot G_0 + P_3 \cdot P_2 \cdot G_1 + G_2 \cdot P_3 + G_3$.

It can be observed from the equations that carry C_{i+1} only depends on the carry C_0 , not on the intermediate carry bits.

Carry-Look-ahead-Adder-Truth-Table

A	B	C_i	C_{i+1}	Condition
0	0	0	0	No carry generate
0	0	1	0	
0	1	0	0	
0	1	1	1	No carry propagate
1	0	0	0	
1	0	1	1	
1	1	0	1	Carry generate
1	1	1	1	

Circuit Diagram

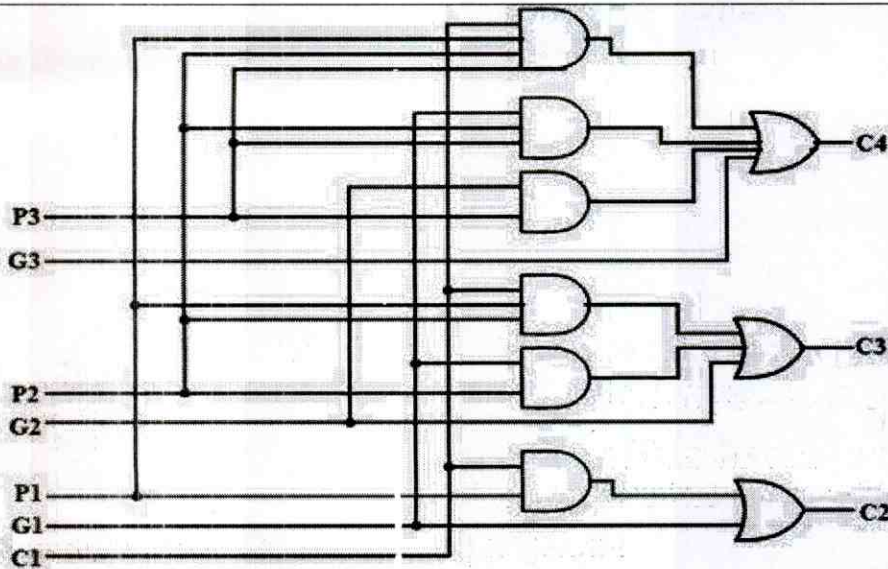
The above equations are implemented using two-level combinational circuits along with AND, OR gates, where gates are assumed to have multiple inputs.



Solution of Question Paper

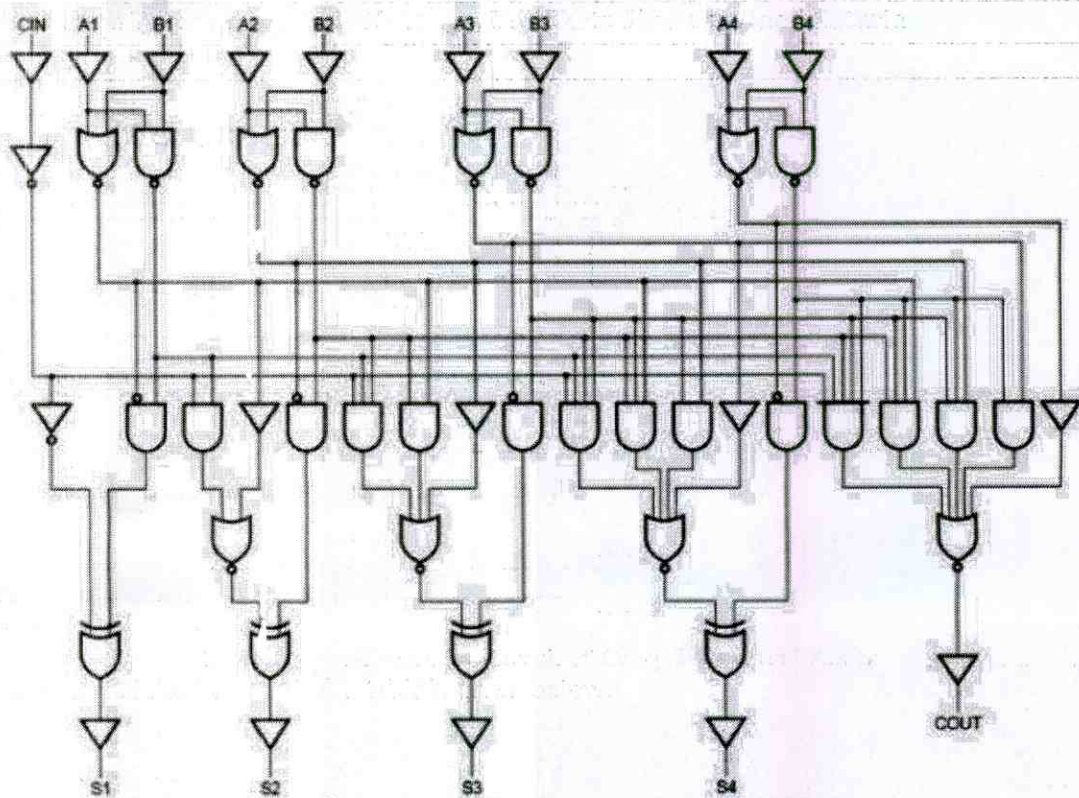
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Carry-Output-Generation-Circuit-of-Carry-Look-ahead-Adder

The Carry Look-ahead Adder circuit for 4-bit is given below.



(4 bit Look Ahead Implementation)



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~~4-bit Carry-Look-ahead Adder Circuit Diagram~~

Q.6 Using JK flip flop design the synchronous counter that counts in the sequence: 001, 100, 010, 101, 110, 111, 011, 001...

Ans. 6

State Table

Present state	Next state	J_1	K_1	J_2	K_2	J_3	K_3
$Q_1 Q_2 Q_3$	$Q_1 Q_2 Q_3$						
1 0 0 1	1 0 0	1	X	0	X	X	1
4 1 0 0	0 1 0	X	1	1	X	0	X
2 0 1 0	1 0 1	1	X	X	1	1	X
5 1 0 1	1 1 0	X	0	1	X	X	1
6 1 1 0	1 1 1	X	0	X	0	1	X
7 1 1 1	0 1 1	X	1	X	0	X	0
3 0 1 1	0 0 1	0	X	X	1	X	0

K-MAP for J_1, K_1

$Q_2 Q_3$	00	01	11	10
Q_1				
0	X	1	0	1
1	X	X	X	X

$$J_1 = \bar{Q}_2 + Q_2 \bar{Q}_3$$

$Q_2 Q_3$	00	01	11	10
Q_1				
0		X	X	X
1	1	0	1	0

$$K_1 = Q_1 \bar{Q}_2 \bar{Q}_3 + Q_2 Q_3$$

K-MAP for J_2, K_2

$Q_2 Q_3$	00	01	11	10
Q_1				
0		0	X	X
1	1	1	X	X

$$J_2 = Q_1$$

$Q_2 Q_3$	00	01	11	10
Q_1				
0		X	1	1
1	X	X	0	0

$$K_2 = \bar{Q}_1 Q_2$$

Solution of Question Paper

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K-MAP for J_3, K_3

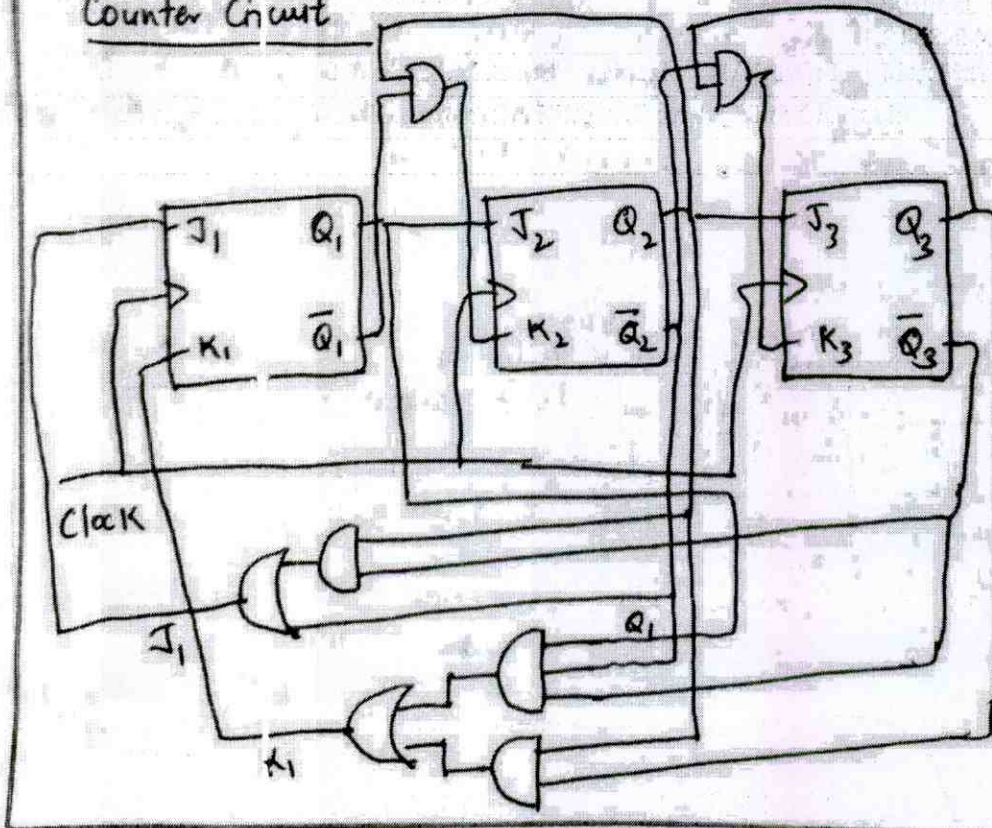
$Q_2 \backslash Q_3$	00	01	11	10
0		X	X	1
1	0	X	X	1

$$J_3 = Q_2$$

$Q_1 \backslash Q_2 Q_3$	00	01	11	10
0		1	0	X
1	X	1	0	X

$$K_3 = \bar{Q}_2 Q_3$$

Counter Circuit





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IIMid Term Examination, December-2022

Semester:	III	Branch:	IT
Subject:	Digital Electronics	Subject Code:	3IT3-04
Time:	1.5 Hours	Maximum Marks:	20
Session : I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q.1 Explain Race around condition
- Q.2 Differentiate between Combinational and sequential circuits
- Q.3 Design a mod 7 asynchronous counter.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q.4 Explain Serial in Parallel out shift register (SIPO) with suitable diagram.
Draw the table indicating action of shift register with every clock.
- Q.5 Draw the truth table, K Map and logic diagram of binary to gray code converter.
- Q.6 Describe the characteristics of Logic Families.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question) (1*6=6)

- Q.7 Implement the following function using 4:1 Multiplexer.
 $F(A, B, C) = \sum m(0, 1, 4, 5, 7)$ Using A and C as selection lines.
- Q.8 Realize the given function Using Tabulation Method.
 $F(A, B, C, D) = \sum m(1, 2, 3, 7, 8, 9, 10, 11, 14, 15)$



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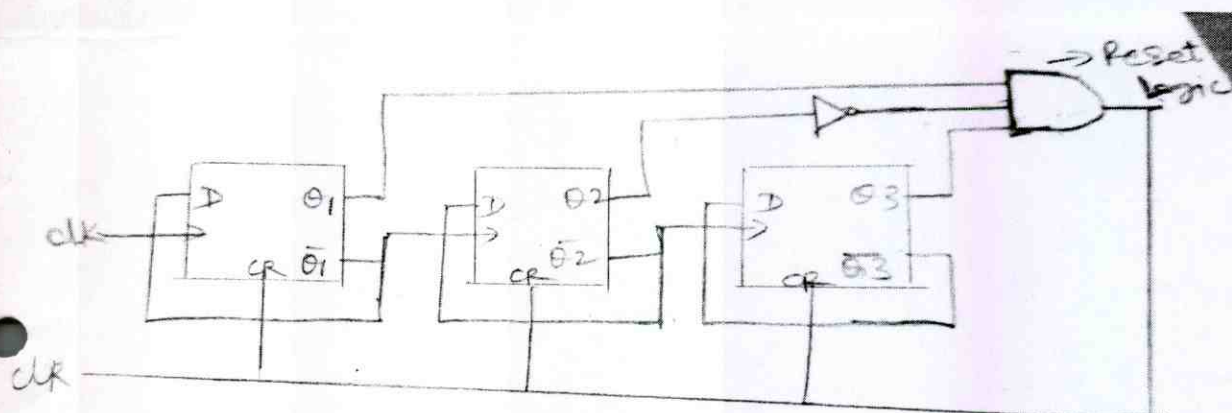
Branch/Semester: IT/III	Subject: Digital Electronics	Subject Code: 3IT3-04
Duration: 1.5 hours	Date: 21/12/22 Session (I/II/III): I	Max Marks: 20
Submitted By: Dr. Sunita Gupta		

Q.1 Race Around Condition In JK Flip-flop – For J-K flip-flop, if $J=K=1$, and if $clk=1$ for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain. This problem is called race around condition in J-K flip-flop.

Q.2 The combinational circuit is time-independent. The output it generates does not depend on any of its previous inputs. On the other hand, sequential circuits are the ones that depend on clock cycles. They depend entirely on the past as well as the present inputs for generating output.

Q.3

In mod-6, there are 6 states. So 3 flipflops are required. It will count from 000 to 101. means from (0 to 5). and then again 000 to 101.



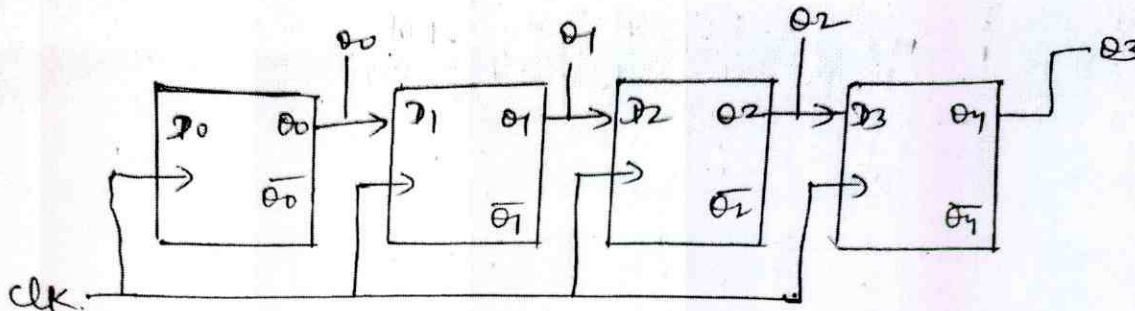
Q.4 SISO → serial in parallel out shift register.

clock	Dis	ff 0 1 2 3
0		0 0 0 0
1	1	1 0 0 0
2	0	0 1 0 0
3	1	1 0 1 0
4	0	0 1 0 1

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Q.5 Binary to Gray code converter

$B_3 B_2 B_1 B_0$	$G_3 G_2 G_1 G_0$
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

$B_3 B_2$	B_1	B_0
00	0	0
01	1	0
11	1	1
10	1	0

$$G_3 = B_3$$

$B_3 B_2$	B_1	B_0
00	0	0
01	1	0
11	1	1
10	1	0

$$G_2 = B_3 \oplus B_2$$

$B_3 B_2$	B_1	B_0
00	0	0
01	1	0
11	1	1
10	1	0

$$G_1 = B_2 \oplus B_1$$

$B_3 B_2$	B_1	B_0
00	0	0
01	1	0
11	1	1
10	1	0

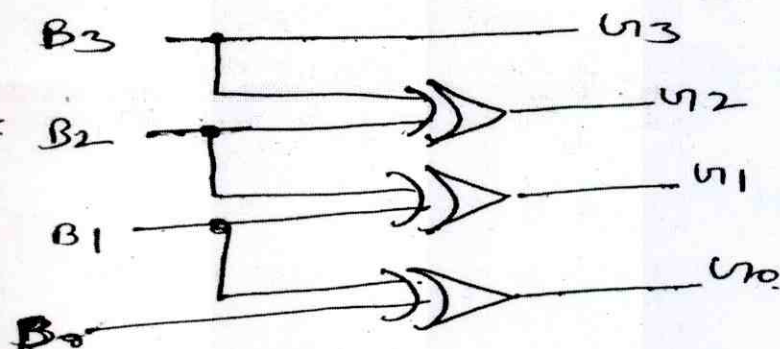
$$G_0 = B_2 \oplus B_1$$



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Binary to Gray code Converter

Q.6

- (1) Speed of operation or propagation delay - it is the time interval between application of an input pulse and occurrence of resulting output pulse.
- (2) Power dissipation - power to operate
- (3) figure of merit - Propagation delay $\times$ Power
- (4) fan-out - maximum no. of loads that output of a gate can drive without impairing its normal operation
- (5) Noise immunity - Ability to tolerate noise signals.
- (6) operating temperature range -
- (7) supply power requirement
- (8) flexibilities available
- (9) source and sinking of current
- (10) current & voltage parameters - $V_{IH}(\min)$, $V_{IL}(\max)$, $V_{OH}(\min)$, $V_{OL}(\max)$, I_{IH} , I_{IL} , I_{OH} , I_{OL} .



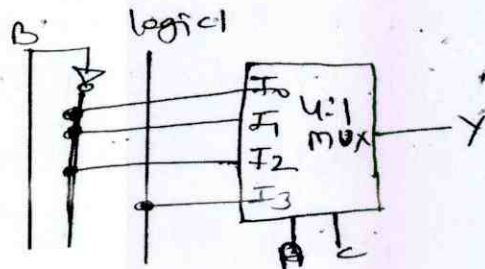
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Q.7

$n=3 \Rightarrow 2^3 = 8:1 \text{ mux}$, But using $4:1 \text{ mux}$, we use mt. method

	I_0	I_1	I_2	I_3
$\bar{B}$	X	X	X	X
B	2	3	6	X
$\bar{B}$	$\bar{B}$	$\bar{B}$	$\bar{B}$	1



Q.8

No. of
1's

0001 (1) ✓	00X1 (1,3) ✓	X0X1 (1,3,9,11) ✓
0010 (2) ✓	X001 (1,9) ✓	X01X (3,3,10,11) ✓
1000 (8) ✓	001X (2,3) ✓	X1X0 (8,9,10,11) ✓
	X010 (2,10) ✓	
0011 (3) ✓	100X (8,9) ✓	XX11 (3,7,11,15) ✓
1001 (9) ✓	10X0 (8,10) ✓	1X1X (10,11,14,15) ✓
1010 (10) ✓		
0111 (7) ✓	0X11 (3,7) ✓	
1011 (11) ✓	X011 (3,11) ✓	
1110 (14) ✓	10X1 (9,11) ✓	
	101X (10,11) ✓	
1111 (15) ✓	1X10 (10,14) ✓	
	X111 (7,15) ✓	
	1X11 (11,15) ✓	
	111X (14,15) ✓	



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Prime implicants	1	2	3	7	8	9	10	11	14	15
$\phi(1, 3, 9, 11)$	X		X			X		X		
$\phi(2, 3, 10, 11)$		X	X				X	X		
$\phi(8, 9, 10, 11)$					X	X	X	X		
$\phi(3, 7, 11, 15)$			X	X				X		X
$\phi(10, 11, 14, 15)$							X	X	X	X
	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

$$Y = \overline{B}D + \overline{B}C + A\overline{B} + CD + AC$$



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II Mid Term Examination, 2022-23

B.Tech./ Semester - III

Branch: CSE/AI/DS/IT

Subject: Data Structures and Algorithms

Subject Code: 3CS4-05/3CAI4-05/3AID4-05/3IT4-05

Time: 1½ Hours

Maximum Marks: 20

Part-I

Note: Attempt All Questions. Each question carries 2 marks (3*2=6)

1. Define Complete Binary Tree?
2. Differentiate between Linear Search and Binary Search.
3. Explain the concept of Minimum Spanning Tree?

Part-II

Note: Attempt any two. Each question carries 4 marks (2*4=8)

4. Write an algorithm for Quick Sort with example.
5. What do you understand by Binary Search Tree? Construct a binary search tree where input list is:
J, R, D, G, T, E, M, H, P, A, F, Q
6. The In-order & Pre-order traversal sequence of nodes in binary tree are given below:
In-order: E X M B S A P T N W H C
Pre-Order: A B X E M S W T P N C H
Draw the binary tree and Write down the algorithm for both the Traversals.

Part-III

Note: Attempt any one question each carries 6 marks.(1*6=6)

7. Explain AVL Tree? Insert the following list of element in an AVL tree: 21, 26, 30, 9, 4, 14, 28, 18, 15
8. Briefly discuss Graph? Write an Algorithm for DFS with example.



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Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CAI4-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 21.12.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

PART-A

1. Define Complete Binary Tree?

Ans. A complete binary tree is a special type of binary tree where all the levels of the tree are filled completely except the lowest level nodes which are filled from as left as possible.

- A complete binary tree is said to be a proper binary tree where all leaves have the same depth.
- In a complete binary tree number of nodes at depth d is 2^d .
- In a complete binary tree with n nodes height of the tree is $\log(n+1)$.
- All the levels except the last level are completely full.

2. Differentiate between Linear Search and Binary Search.

Ans.

Basis of comparison	Linear search	Binary search
Definition	The linear search starts searching from the first element and compares each element with a searched element till the element is not found.	It finds the position of the searched element by finding the middle element of the array.
Sorted data	In a linear search, the elements don't need to be arranged in sorted order.	The pre-condition for the binary search is that the elements must be arranged in a sorted order.
Implementation	The linear search can be implemented on any linear data structure such as an array, linked list, etc.	The implementation of binary search is limited as it can be implemented only on those data structures that have two-way traversal.
Approach	It is based on the sequential approach.	It is based on the divide and conquer approach.
Size	It is preferable for the small-sized data sets.	It is preferable for the large-size data sets.



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Question Paper Solution

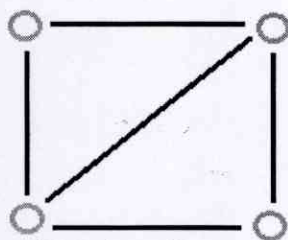
Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

3. Explain the concept of Minimum Spanning Tree?

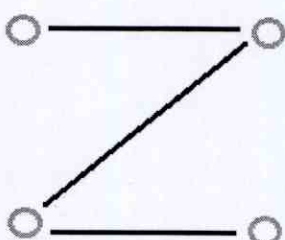
Ans. A minimum spanning tree is a special kind of tree that minimizes the lengths (or "weights") of the edges of the tree. An example is a cable company wanting to lay line to multiple neighborhoods; by minimizing the amount of cable laid, the cable company will save money.

A tree has one path joins any two vertices. A spanning tree of a graph is a tree that:

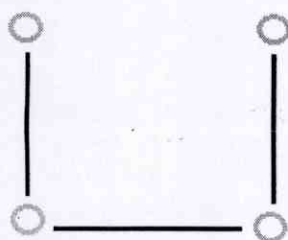
- Contains all the original graph's vertices.
- Reaches out to (spans) all vertices.
- Is acyclic. In other words, the graph doesn't have any nodes which loop back to itself.



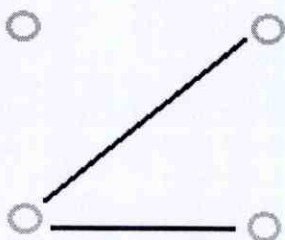
Not Acyclic



Acyclic



Connected



Not Connected



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Question Paper Solution

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Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

PART-B

4. Write an algorithm for Quick Sort with example.

Ans.

QUICKSORT (array A, start, end)

```
{  
1 if (start < end)  
2 {  
3 p = partition(A, start, end)  
4 QUICKSORT (A, start, p - 1)  
5 QUICKSORT (A, p + 1, end)  
6 }  
}
```

Partition Algorithm:

The partition algorithm rearranges the sub-arrays in a place.

PARTITION (array A, start, end)

```
{  
1 pivot = A[end]  
2 i = start-1  
3 for j = start to end -1 {  
4 do if (A[j] < pivot) {  
5 then i = i + 1  
6 swap A[i] with A[j]  
7 }}  
8 swap A[i+1] with A[end]  
9 return i+1  
}
```




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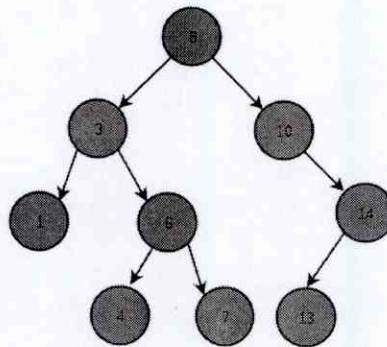
Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

5. What do you understand by Binary Search Tree? Construct a binary search tree where input list is:
J, R, D, G, T, E, M, H, P, A, F, Q

Ans. Binary Search Tree is a node-based binary tree data structure which has the following properties:

- The left subtree of a node contains only nodes with keys lesser than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- The left and right subtree each must also be a binary search tree.

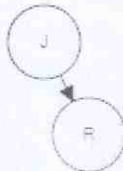


Construct a binary search tree where input list is:
J, R, D, G, T, E, M, H, P, A, F, Q

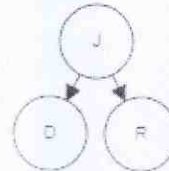
Insert J



Insert R



Insert D





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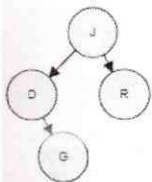
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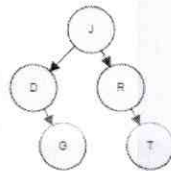
Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
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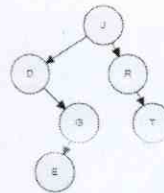
Insert G



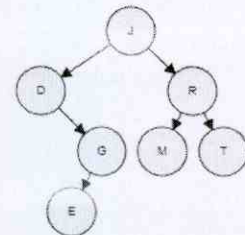
Insert T



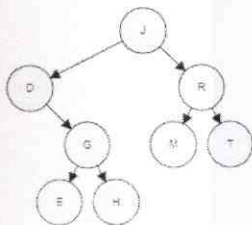
Insert E



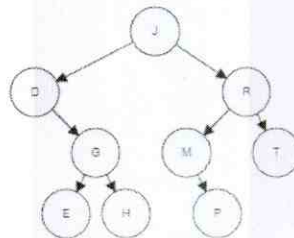
Insert M



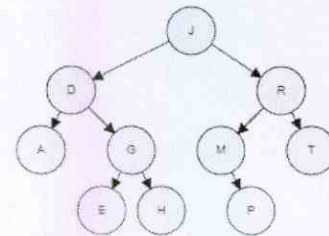
Insert H



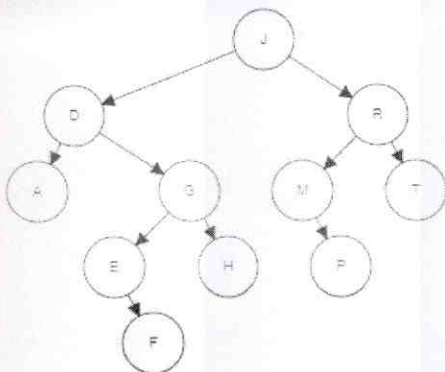
Insert P



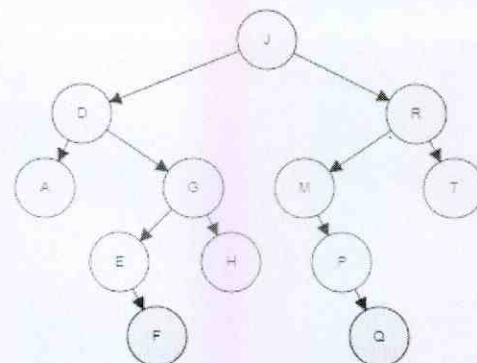
Insert A



Insert F



Insert Q



Final Binary Search tree



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Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

6. The In-order & Pre-order traversal sequence of nodes in binary tree are given below:

In-order: E X M B S A P T N W H C

Pre-Order: A B X E M S W T P N C H

Draw the binary tree.

Answer.

Algorithms

InOrder (T)

```
If T <> null
print ('empty tree')
If T.lp <> null
then InOrder(T.lp)
print (T.data)
If T.rp <> null
then InOrder (T.rp)
end.
```

PreOrder (T)

```
If T <> Null
then print (T.data)
else print('empty tree')
If T.lp <> null
then PreOrder(T.lp)
If T.rp <> null
then preorder (T.rp)
end.
```



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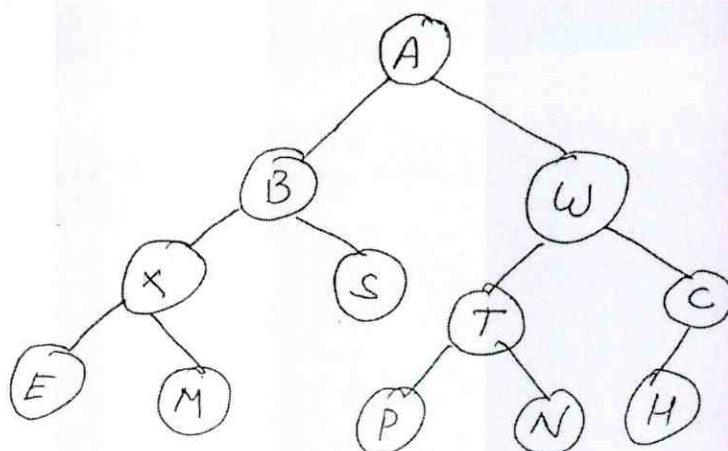
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Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks: 20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

Inorder :- E X M B S A P T N W H C
Pre-order :- A B X E M S W T P N C H





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Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

PART-C

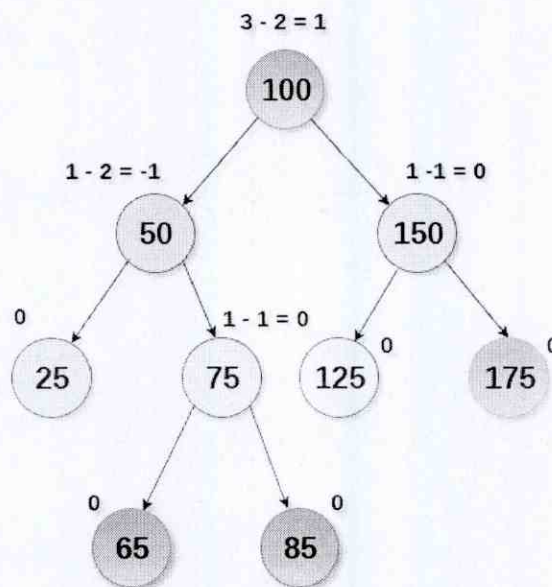
7. Explain AVL Tree? Insert the following list of element in an AVL tree: 10, 20, 30, 35, 25, 5, 40, 45, 34
Ans.

AVL tree is a self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees cannot be more than one for all nodes.

AVL Tree can be defined as height balanced binary search tree in which each node is associated with a balance factor which is calculated by subtracting the height of its right sub-tree from that of its left sub-tree.

Tree is said to be balanced if balance factor of each node is in between -1 to 1, otherwise, the tree will be unbalanced and need to be balanced.

Balance Factor (k) = height (left(k)) - height (right(k))



AVL Tree



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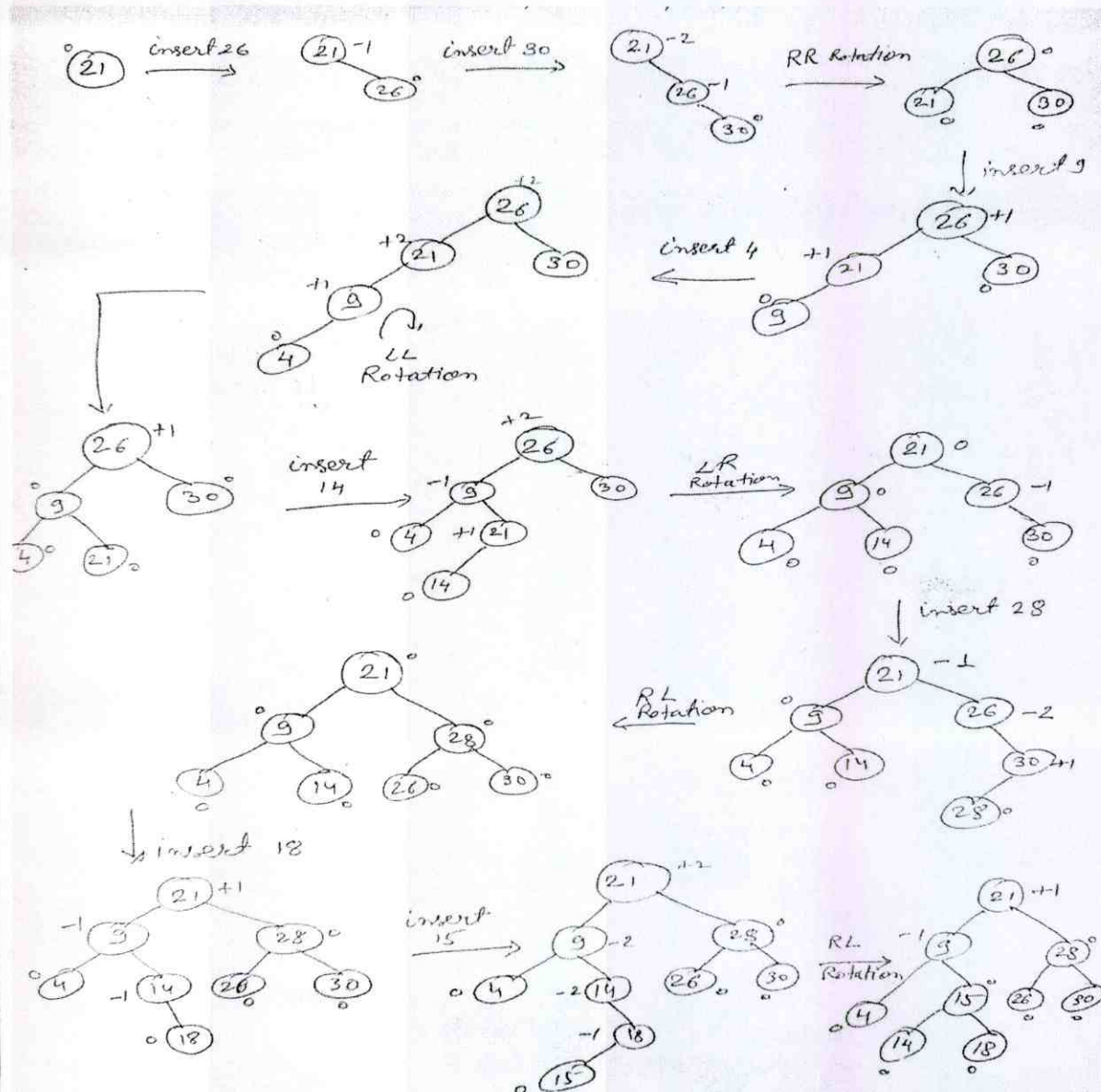
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Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

AVL Tree Creation: - 21, 26, 30, 9, 4, 14, 28, 18, 15





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Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Rajesh Rajaan		

8. Briefly discuss Graph? Write an Algorithm for DFS with example.

Ans. A graph can be defined as group of vertices and edges that are used to connect these vertices. A graph can be seen as a cyclic tree, where the vertices (Nodes) maintain any complex relationship among them instead of having parent child relationship.

A graph G can be defined as an ordered set $G(V, E)$ where $V(G)$ represents the set of vertices and $E(G)$ represents the set of edges which are used to connect these vertices.

Step 1: SET STATUS = 1 (ready state) for each node in G

Step 2: Push the starting node A on the stack and set its STATUS = 2 (waiting state)

Step 3: Repeat Steps 4 and 5 until STACK is empty

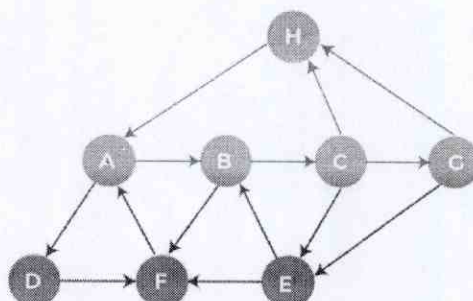
Step 4: Pop the top node N . Process it and set its STATUS = 3 (processed state)

Step 5: Push on the stack all the neighbors of N that are in the ready state (whose STATUS = 1) and set their STATUS = 2 (waiting state)

[END OF LOOP]

Step 6: EXIT

Example:-



Adjacency Lists

A : B, D
B : C, F
C : E, G, H
D : F
E : B, F
F : A
G : F
H : A



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Question Paper Solution

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Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

Now, let's start examining the graph starting from Node H.

Step 1 - First, push H onto the stack.

STACK: H

Step 2 - POP the top element from the stack, i.e., H, and print it. Now, PUSH all the neighbors of H onto the stack that are in ready state.

Print: H STACK: A

Step 3 - POP the top element from the stack, i.e., A, and print it. Now, PUSH all the neighbors of A onto the stack that are in ready state.

Print: A STACK: B, D

Step 4 - POP the top element from the stack, i.e., D, and print it. Now, PUSH all the neighbors of D onto the stack that are in ready state.

Print: D STACK: B, F

Step 5 - POP the top element from the stack, i.e., F, and print it. Now, PUSH all the neighbors of F onto the stack that are in ready state.

Print: F STACK: B

Step 6 - POP the top element from the stack, i.e., B, and print it. Now, PUSH all the neighbors of B onto the stack that are in ready state.

Print: B STACK: C

Step 7 - POP the top element from the stack, i.e., C, and print it. Now, PUSH all the neighbors of C onto the stack that are in ready state.

Print: C STACK: E, G



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Question Paper Solution

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Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Rajaan		

Step 8 - POP the top element from the stack, i.e., G and PUSH all the neighbors of G onto the stack that are in ready state.

Print: G STACK: E

Step 9 - POP the top element from the stack, i.e., E and PUSH all the neighbors of E onto the stack that are in ready state.

Print: E STACK:



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I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	CSE/CS(AI)/CS(DS)/IT
Subject:	Object Oriented Programming	Subject Code:	3CS4-06/3CAI4-06/3IT4-06
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. List all the characteristics of object oriented paradigm.
- Q2. What are the methods of defining the class member function? List all
- Q3. Define the role of public and private access specifiers within a class.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Define classes and object? Explain the process of creating and destroying objects dynamically using "new" and "delete" operators with the help of example.
- Q5. What do you mean by function overloading? Explain with example.
- Q6. Write the difference between constructor and destructor.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q7. Write a C++ program to display student details (Name, Roll no, Branch, contact) using constructor (default, copy, parameterized)
- Q8. Develop a program which add two complex numbers and return an object from function.



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Solution of Question Paper
II Mid-Term Examination, DEC. -2022

Branch/Semester: CSE/CS(AI)/CS(DS)/IT/III	Subject: Object Oriented Programming	Subject Code: 3CS4-06/3CAI4-06/3IT4-06/3CDS4-06
Duration: 1.5 hours	Date: 19/12/2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shalini Singhal, Garima Gupta, Ashish Pant, M.K Beniwal, Priya Gupta, Dolly Mittal, Kailash Soni		

PART A (short-answer type questions)

Ans.1 A syntax error is something caught by the compiler/interpreter and its incorrect use of the language itself. For example, for which is invalid Python.

A runtime error is a problem that cannot be detected before the code runs but causes an issue that is caught during the program run.

A runtime error is an error that occurs while running a computer program while a logical error is an error in a program that causes it to operate incorrectly, but not to terminate abnormally. This is the main difference between runtime error and logical error.

Ans.2 A pure virtual function is a function that must be overridden in a derived class and need not be defined. A virtual function is declared to be "pure" using the curious =0 syntax. For example:

```
class Base {  
    class Vehicle {  
    public:  
        // performs the operation  
        virtual void input= 0  
    };  
};
```

Ans.3. Inheritance is the procedure in which one class inherits the attributes and methods of another class. The class whose properties and methods are inherited is known as the Parent class. And the class that inherits the properties from the parent class is the Child class.

- Single Inheritance.
- Multiple Inheritance.
- Multilevel Inheritance.
- Hierarchical Inheritance.
- Hybrid Inheritance.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Ans 4. Exception handling is the process of responding to unwanted or unexpected events when a computer program runs. Exception handling deals with these events to avoid the program or system crashing, and without this process, exceptions would disrupt the normal operation of a program.

It can consist of 3 steps:

1. a try block that encloses the code section which might throw an exception,
2. one or more catch blocks that handle the exception and.
3. a finally block which gets executed after the try block was successfully executed or a thrown exception was handled.

```
#include <iostream>
```



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II Mid-Term Examination, DEC. -2022

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Duration: 1.5 hours	Date: 19/12/2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shalini Singhal, Garima Gupta, Ashish Pant, M.K Beniwal, Priya Gupta, Dolly Mittal, Kailash Soni		

```
using namespace std;

double division(int a, int b) {
    if( b == 0 ) {
        throw "Division by zero condition!";
    }
    return (a/b);
}

int main () {
    int x = 50;
    int y = 0;
    double z = 0;

    try {
        z = division(x, y);
        cout<< z <<endl;
    } catch (const char* msg) {
        cerr<<msg<<endl;
    }

    return 0;
}
```

Q.5// C++ program to implement function templates
#include <iostream>
using namespace std;

```
// Function template to swap
// two numbers
template <class T>
int swap_numbers(T& x, T& y)
{
    T t;
    t = x;
    x = y;
    y = t;
    return 0;
}

// Driver code
int main()
{
    int a, b;
    a = 10, b = 20;
    // Invoking the swap()
```




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```
swap_numbers(a, b);  
cout<< a << " " << b << endl;  
return 0;  
}
```

Output:

20 10

Ans.6 Dynamic binding or late binding is the mechanism a computer program waits until runtime to bind the name of a method called to an actual subroutine. It is an alternative to early binding or static binding where this process is performed at compile-time.

Dynamic binding in C++ is a practice of connecting the function calls with the function definitions by avoiding the issues with static binding, which occurred at build time. Because dynamic binding is flexible, it avoids the drawbacks of static binding, which connected the function call and definition at build time.

```
#include <iostream>  
using namespace std;  
class A {  
public:  
    void final_print() // function that call display  
    {  
        display ();  
    }  
    virtual void display () // the display function  
    {  
        cout<< "Printing from the base class" << endl;  
    }  
};  
class B : public A // B inherit a publicly  
{  
public:  
    virtual void display () // B's display  
    {  
        cout<< "Printing from the derived class" << endl;  
    }  
};  
int main()  
{  
    A obj1; // Creating A's pbject  
    obj1.final_print (); // Calling final_print  
    B obj2; // calling b  
    obj2.final_print ();  
    return 0; }  
}
```

Output: -

Printing from the base class



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Printing from the derived class

PART C (Descriptive/Analytical/Problem solving/Design questions)
(Attempt any 1 Question) (1*6=6)

Ans.7

```
C++
#include<iostream.h>
#include<conio.h>
#include<fstream.h>
#include<stdlib.h>
#include<iomanip.h>
#include<string.h>
#include<stdio.h>
const char* fileName="Employee.txt";

class Employee
{
private:
int EmpID;
char EmpName[50],Post[50],Department[10];
float Salary;
public:
void ReadData();
int GetID();
void DisplayRecord();
char* GetDepartment();
};

void Employee::ReadData()
{
cout<<endl<<"Employee ID:";
cin>>EmpID;
cout<<"Employee Name:";
cin>>EmpName;
cout<<"Employee's Post:";
cin>>Post;
cout<<"Employee's Department:";
cin>>Department;
cout<<"Salary:";
cin>>Salary;
}
```




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```
void Employee::DisplayRecord()
{
    cout<<endl<<"_____";
    cout<<endl<<setw(5)<<EmpID<<setw(15)<<EmpName<<setw(15)<<Post<<setw(15)<<Department<<setw(8)<<Salary;
}

int Employee::GetID()
{
    return EmpID;
}

char* Employee::GetDepartment()
{
    return Department;
}

void main()
{
    Employee emp,e;
    char option,ch,Dept[50];
    int ID,isFound;
    clrscr();
    fstreamfile;
    file.open(fileName,ios::ate|ios::in|ios::out|ios::binary);
    do
    {
        cout<<"*****Menu*****";
        cout<<endl<<"Enter your option";
        cout<<endl<<"1 => Add a new record";
        cout<<endl<<"2 => Search record from employee id";
        cout<<endl<<"3 => List Employee of particular department";
        cout<<endl<<"4 => Display all employee";
        cout<<endl<<"5 => Update record of an employee";
        cout<<endl<<"6 => Delete record of particular employee";
        cout<<endl<<"7 => Exit from the program"<<endl;
        cout<<"*****"<<endl;
        cin>>option;
        switch(option)
        {
            case '1':
                emp.ReadData();
                file.seekg(0,ios::beg);
                isFound=0;
                file.read((char*)&e,sizeof(e));
                while(!file.eof())
```



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```
{
if(e.GetID()==emp.GetID())
{
cout<<"This ID already exist...Try for another ID";
isFound=1;
break;
}
file.read((char*)&e,sizeof(e));
}
if(isFound==1)
break;
file.clear();
file.seekp(0,ios::end);
file.write((char*)&emp, sizeof(emp));
cout<<endl<<"New record has been added successfully...";
break;

case '2':
isFound=0;
cout<<endl<<"Enter ID of an employee to be searched:";
cin>>ID;
file.seekg(0,ios::beg);
file.read((char*)&e,sizeof(e));
while(!file.eof())
{
if(e.GetID()==ID)
{
cout<<endl<<"The record found...."<<endl;
cout<<endl<<setw(5)<<"ID"<<setw(15)<<"Name"<<setw(15)<<"Post"<<setw(15)<<"Department"<<setw(8)<<"Salary";
c.DisplayRecord();
isFound=1;
break;
}
file.read((char*)&e,sizeof(e));
}
file.clear();
if(isFound==0)
cout<<endl<<"Data not found for employee ID#"<<ID;
break;
case '3':
isFound=0;
cout<<"Enter department name to list employee within it:";
cin>>Dept;
file.seekg(0,ios::beg);
file.read((char*)&e,sizeof(e));
```




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Submitted By: Shalini Singhal, Garima Gupta, Ashish Pant, M.K Beniwal, Priya Gupta, Dolly Mittal, Kailash Soni		

```
while(!file.eof())
{
if(strcmp(e.GetDepartment(),Dept)==0)
{
cout<<endl<<"The record found for this department"<<endl;

cout<<endl<<setw(5)<<"ID"<<setw(15)<<"Name"<<setw(15)<<"Post"<<setw(15)<<"Department"<<setw(8)<<"Salary";
e.DisplayRecord();
isFound=1;
break;
}
file.read((char*)&e,sizeof(e));
}
file.clear();
if(isFound==0)
cout<<endl<<"Data not found for department"<<Dept;
break;

case '4':
cout<<endl<<"Record for employee";
file.clear();
file.seekg(0,ios::beg);
int counter=0;
file.read((char*)&e,sizeof(e));
while(!file.eof())
{
counter++;
if(counter==1)
{
cout<<endl<<setw(5)<<"ID"<<setw(15)<<"Name"<<setw(15)<<"Post"<<setw(15)<<"Department"<<setw(8)<<"Salary";
}
e.DisplayRecord();
file.read((char*)&e,sizeof(e));
}
cout<<endl<<counter<<"records found.....";
file.clear();
break;

case '5':
int recordNo=0;
cout<<endl<<"File is being modified....";
cout<<endl<<"Enter employee ID to be updated:";
cin>>ID;
isFound=0;
file.seekg(0,ios::beg);
```



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```
file.read((char*)&e,sizeof(e));
while(!file.eof())
{
    recordNo++;
    if(e.GetID()==ID)
    {
        cout<<"The old record of employee having ID"<<ID<<" is:";
        e.DisplayRecord();
        isFound=1;
        break;
    }
    file.read((char*)&e,sizeof(e));
}

if(isFound==0)
{
    cout<<endl<<"Data not found for employee ID#"<<ID;
    break;
}
file.clear();
int location=(recordNo-1)*sizeof(e);
file.seekp(location,ios::beg);
cout<<endl<<"Enter new record for employee having ID"<<ID;
e.ReadData();
file.write((char*)&e, sizeof(e));
break;

case '6':
    recordNo=0;
    cout<<endl<<"Enter employment ID to be deleted:";
    cin>>ID;
    isFound=0;
    file.seekg(0,ios::beg);
    file.read((char*)&e,sizeof(e));
    while(!file.eof())
    {
        recordNo++;
        if(e.GetID()==ID)
        {
            cout<<" The old record of employee having ID "<<ID<<" is: ";
            e.DisplayRecord();
            isFound=1;
            break;
        }
        file.read((char*)&e,sizeof(e));
    }
```




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Submitted By: Shalini Singhal, Garima Gupta, Ashish Pant, M.K Beniwal, Priya Gupta, Dolly Mittal, Kailash Soni		

```
}
char tempFile[]="temp.txt";
fstream temp(tempFile,ios::out|ios::binary);
if(isFound==0)
{
cout<<endl<<"Data not found for employee ID#"<<ID;
break;
}
else
{
file.clear();
file.seekg(0,ios::beg);
file.read((char*)&e,sizeof(e));
while(!file.eof())
{
if(e.GetID()!=ID)
temp.write((char*)&e,sizeof(e));
file.read((char*)&e,sizeof(e));
}
file.close();
temp.close();
temp.open(tempFile,ios::in|ios::binary);
file.open(fileName,ios::out|ios::binary);
temp.read((char*)&e,sizeof(e));
while(!temp.eof())
{
file.write((char*)&e,sizeof(e));
temp.read((char*)&e,sizeof(e));
}
temp.close();
file.close();
remove(tempFile);
file.open(fileName,ios::ate|ios::in|ios::out|ios::binary);
break;

case '7':
exit(0);
break;

default:
cout<<"Invalid Options";
}
cout<<"\nDo you want to continue.....?y/n";
cin>>ch;
```



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```
}while(ch!='n');  
}
```

Ans.8 Polymorphism in C++ means, the same entity (function or object) behaves differently in different scenarios. Consider this example: The “+” operator in c++ can perform two specific functions at two different scenarios i.e when the “+” operator is used in numbers, it performs addition. Types of Polymorphism in C++

Polymorphism in C++ is categorized into two types. The figure below shows the types:

1. Compile Time Polymorphism

In compile-time polymorphism, a function is called at the time of program compilation. We call this type of polymorphism as early binding or Static binding.

Function overloading and operator overloading is the type of Compile time polymorphism.

1. Function Overloading

Function overloading means one function can perform many tasks. In C++, a single function is used to perform many tasks with the same name and different types of arguments. In the function overloading function will call at the time of program compilation. It is an example of compile-time polymorphism.

In the below example, A function ADD() is used to perform two tasks. The two tasks would be to add two integer values and add two strings (concatenate).

Readability of the program increases by function overloading. It is achieved by using the same name for the same action.

Source Code:

```
1 #include <iostream>  
2 using namespace std;  
3 class Addition {  
4 public:  
5     int ADD(int X, int Y) // Function with parameter  
6     {  
7         return X+Y; // this function is performing addition of two Integer value  
8     }  
9     int ADD() { // Function with same name but without parameter  
10         string a= "HELLO";  
11         string b="SAM"; // in this function concatenation is performed  
12         string c= a+b;  
13         cout<<c<<endl;
```




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```
14
15 }
16};
17int main(void) {
18  Addition obj; // Object is created
19  cout<<obj.ADD(128, 15)<<endl; //first method is called
20  obj.ADD(); // second method is called
21  return 0;
22}
```

Output

143

HELLOSAM

In the above example, we use function ADD() to perform many tasks which is a property of polymorphism in C++.

II. Operator Overloading

Operator overloading means defining additional tasks to operators without changing its actual meaning. We do this by using operator function.

The purpose of operator overloading is to provide a special meaning to the user-defined data types.

The advantage of Operators overloading is to perform different operations on the same operand.

Source Code

```
1 #include <iostream>
2 using namespace std;
3 class A
4 {
5
6  string x;
7  public:
8  A(){}
9  A(string i)
10 {
11  x=i;
12 }
13 void operator+(A);
14 void display();
```



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```
15};
16
17void A::operator+(A a)
18{
19
20    string m = x+a.x;
21    cout<<"The result of the addition of two objects is : "<<m;
22
23}
24int main()
25{
26    A a1("Welcome");
27    A a2("back");
28    a1+a2;
29    return 0;
30}
```

Output

The result of the addition of two objects is: Welcomeback

2. Runtime Polymorphism

In a Runtime polymorphism, functions are called at the time the program execution. Hence, it is known as late binding or dynamic binding.

Function overriding is a part of runtime polymorphism. In function overriding, more than one method has the same name with different types of the parameter list.

It is achieved by using virtual functions and pointers. It provides slow execution as it is known at the run time. Thus, It is more flexible as all the things executed at the run time.

I. Function overriding

In function overriding, we give the new definition to base class function in the derived class. At that time, we can say the base function has been overridden. It can be only possible in the 'derived class'. In function overriding, we have two definitions of the same function, one in the superclass and one in the derived class. The decision about which function definition requires calling happens at **runtime**. That is the reason we call it 'Runtime polymorphism'.

Source code

```
1 #include <iostream>
```




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```
2 using namespace std;
3 class Animal {
4     public:
5     void function() {
6         cout << "Eating..." << endl;
7     }
8 };
9 class Man: public Animal
10 {
11     public:
12     void function()
13     {
14         cout << "Walking ..." << endl;
15     }
16 };
17 int main(void) {
18     Animal A = Animal();
19     A.function(); //parent class object
20     Man m = Man();
21     m.function(); //child class object
22
23     return 0;
24 }
```

Eating

Walking

II. Virtual Function

A virtual function is declared by keyword virtual. The return type of virtual function may be int, float, void. A virtual function is a member function in the base class. We can redefine it in a derived class. It is part of run time polymorphism. The declaration of the virtual function must be in the base class by using the keyword virtual. A virtual function is not static.

The virtual function helps to tell the compiler to perform dynamic binding or late binding on the function. If it is necessary to use a single pointer to refer to all the different classes' objects. This is because we will have to create a pointer to the base class that refers to all the derived objects.



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But, when the base class pointer contains the derived class address, the object always executes the base class function. For resolving this problem, we use the virtual function. When we declare a virtual function, the compiler determines which function to invoke at runtime. Let's see the below example for understanding how the program execution happens without virtual function and with virtual function.

Source code

```
1
2
3
4 without virtual Function
5 #include <iostream>
6 using namespace std;
7 class Add
8 {
9     int x=5, y=20;
10    public:
11    void display() //overridden function
12    {
13        cout<< "Value of x is : " <<x+y<<endl;
14    } };
15 class Subtract: public Add
16 {
17     int y = 10, z=30;
18     public:
19     void display() //overridden function
20     {
21         cout<< "Value of y is : " <<y-z<<endl;
22     }
23 };
24 int main()
25 {
26     Add *m;        //base class pointer .it can only access the base class members
27     Subtract s;    // making object of derived class
28     m = &s;
29     m->display();  // Accessing the function by using base class pointer
30     return 0;
```

Output

Value of x is: 25



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II Mid Term Examination, Dec-2022**

Semester:	III	Branch:	CS/DS/AI/IT
Subject:	Software Engineering	Subject Code:	3CS4-07/3AID4-07/3CDS4-07/3IT4-07
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III):II			

PART A

(Short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. List the requirement analysis tasks.
Q2. Why accuracy is important in Data Dictionary?
Q3. Briefly describe Object-oriented design approaches

PART B

(Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Explain FSM with a proper Example.
Q5. Describe different design principles.
Q6. Describe Object-Oriented Process model with a neat diagram.

PART C

(Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q7. Draw Sequence Diagram for withdrawing cash from ATM.
Q8. Define Effective Modular Design. Differentiate between Cohesion and Coupling in detail.



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Question Paper Solution
Mid-Term II
2022-23

Branch: CS/IT/AI/DS

Semester: III

Subject: SE

Submitted By:- Dr.Rajat Goel, Ms.Aakriti Sharma, Ms. Mamta Sakpal, Ms. Alisha Goyal, Ms. Priyanka Bharadwaj,
Ms Sanju Choudhary

Q1. List the requirement analysis tasks.

Ans. The tasks are:

- Problem Recognition
- Evaluation and Synthesis
- Modeling
- Specification
- Review

Q2. Why accuracy is important in Data Dictionary?

Ans. Data dictionary contains data about data. It should have accurate information of the name, type, length, property, reason of use, scope etc. This information is crucial for the team members engaged in later phases from coding to maintenance.

Q3. Briefly describe Object-oriented design approaches.

Ans. The different approaches are:

- The Booch method: The Booch method encompasses both a "micro development process" and a "macro development process." Macro development encompasses an architectural planning activity while Micro development defines a set of "rules" that govern the use of operations and attributes
- The Rumbaugh method: It encourages design to be conducted at two different levels of abstraction i.e. System design and Object design.
- The Jacobson method: It emphasizes traceability to the OOSE analysis model.
- The Coad and Yourdon method: The approach addresses the infrastructure for the application and representation of major system components.
- The Wirfs-Brock method: In this method Protocols for each class are constructed by refining contracts between objects.

Q4. Explain FSM with a proper Example.

Ans. Finite state machine is used to recognize patterns. Finite automata machine takes the string of symbol as input and changes its state accordingly. In the input, when a desired symbol is found then the transition occurs. While transition, the automata can either move to the next state or stay in the same state. FA has two states: accept state or reject state. When the input string is successfully processed and the automata reached its final state then it will accept. A finite automata consists of following:

Q: finite set of states

Σ : finite set of input symbol

q0: initial state

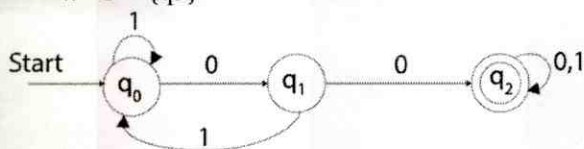
F: final state

δ : Transition function

Transition function can be define as

$\delta: Q \times \Sigma \rightarrow Q$

1. $Q = \{q_0, q_1, q_2\}$
2. $\Sigma = \{0, 1\}$
3. $q_0 = \{q_0\}$
4. $F = \{q_2\}$



Q5. Describe different design principles.

Ans. The principles are:

- The design process should not suffer from "tunnel vision." A good designer should consider alternative approaches, judging each based on the requirements of the problem, the resources available to do the job, and the design concepts.
- The design should be traceable to the analysis model. Because a single element of the design model often traces to multiple requirements, it is necessary to have a means for tracking how requirements have been satisfied by the design model.



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Question Paper Solution
Mid-Term II
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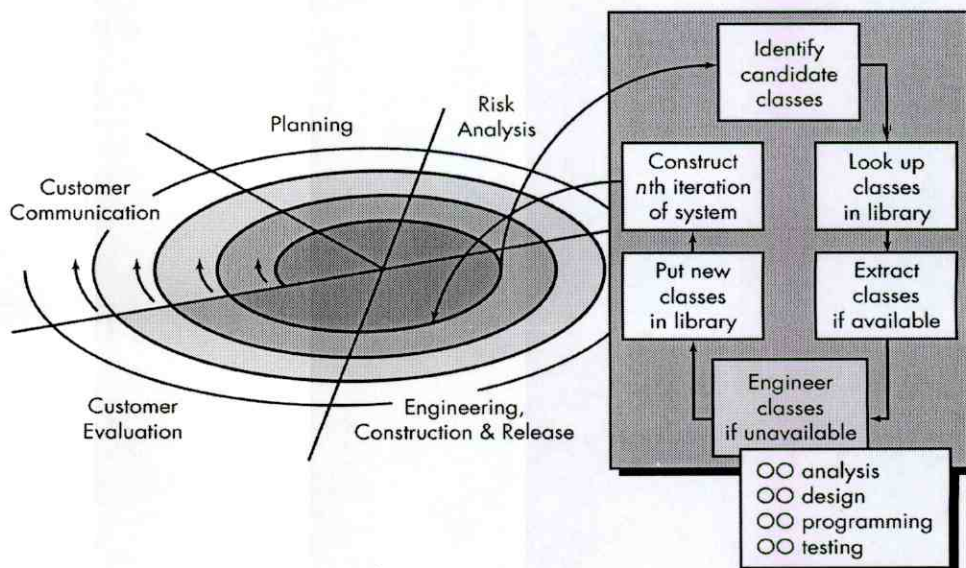
Subject: SE

Submitted By:- Dr.Rajat Goel, Ms.Aakriti Sharma, Ms. Mamta Sakpal, Ms. Alisha Goyal , Ms. Priyanka Bharadwaj,
Ms Sanju Choudhary

- The design should not reinvent the wheel. Time is short and resources are limited! Design time should be invested in representing truly new ideas and integrating those patterns that already exist.
- The design should “minimize the intellectual distance” between the software and the problem as it exists in the real world. That is, the structure of the software design should (whenever possible) mimic the structure of the problem domain.
- The design should exhibit uniformity and integration. A design is uniform if it appears that one person developed the entire thing. Rules of style and format should be defined for a design team before design work begins. A design is integrated if care is taken in defining interfaces between design components.
- The design should be structured to accommodate change.
- The design should be structured to degrade gently, even when aberrant data, events, or operating conditions are encountered.
- Design is not coding; coding is not design. Even when detailed procedural designs are created for program components, the level of abstraction of the design model is higher than source code. The only design decisions made at the coding level address the small implementation details that enable the procedural design to be coded.
- The design should be assessed for quality as it is being created, not after the fact. A variety of design concepts and design measures are available to assist the designer in assessing quality.
- The design should be reviewed to minimize conceptual (semantic) errors. There is sometimes a tendency to focus on minutiae when the design is reviewed, missing the forest for the trees. A design team should ensure that major conceptual elements of the design (omissions, ambiguity, inconsistency) have been addressed before worrying about the syntax of the design model.

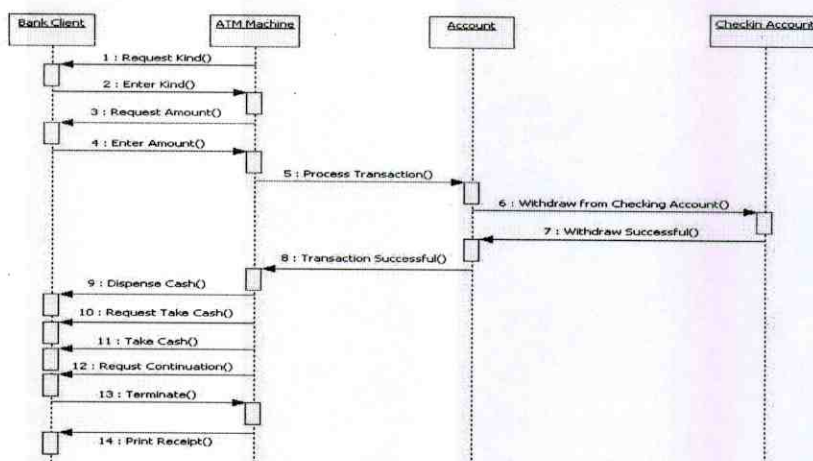
Q6. Describe Object-Oriented Process model with a neat diagram.

Ans. The OO process moves through an evolutionary spiral that starts with customer communication. It is here that the problem domain is defined and that basic problem classes (discussed later in this chapter) are identified. Planning and risk analysis establish a foundation for the OO project plan. The technical work associated with OO software engineering follows the iterative path shown in the shaded box. OO software engineering emphasizes reuse. Therefore, classes are “looked up” in a library (of existing OO classes) before they are built. When a class cannot be found in the library, the software engineer applies object-oriented analysis (OOA), object-oriented design (OOD), object-oriented programming (OOP), and object-oriented testing (OOT) to create the class and the objects derived from the class. The new class is then put into the library so that it may be reused in the future. The object-oriented view demands an evolutionary approach to software engineering. It would be exceedingly difficult to define all necessary classes for a major system or product in a single iteration. As the OO analysis and design models evolve, the need for additional classes becomes apparent. It is for this reason that the paradigm just described works best for OO.



Q7. Draw Sequence Diagram for withdrawing cash from ATM.

Ans.



Q8. Define Effective Modular Design. Differentiate between Cohesion and Coupling in detail.

Ans. A modular design reduces complexity, facilitates change (a critical aspect of software maintainability), and results in easier implementation by encouraging parallel development of different parts of a system. The concept of functional independence is a direct outgrowth of modularity and the concepts of abstraction and information hiding. Functional independence is achieved by developing modules with "single-minded" function and an "aversion" to excessive interaction with other modules. Stated another way, we want to design software so that each module addresses a specific subfunction of requirements and has a simple interface when viewed from other parts of the program structure. It is fair to ask why independence is important. Software with effective modularity, that is, independent modules, is easier to develop because function may be compartmentalized and interfaces are simplified (consider the ramifications when development is conducted by a team). Independent modules are easier to maintain (and test) because secondary effects caused by design or code modification are limited, error propagation is reduced, and reusable modules are possible. To summarize, functional independence is a key to good design, and design is the key to software quality. Independence is measured using two qualitative criteria: cohesion and coupling.

Coupling	Cohesion
Coupling is also called Inter-Module Binding.	Cohesion is also called Intra-Module Binding.
Coupling shows the relationships between modules.	Cohesion shows the relationship within the module.
Coupling shows the relative independence between the modules.	Cohesion shows the module's relative functional strength.
While creating, you should aim for low coupling, i.e., dependency among modules should be less.	While creating you should aim for high cohesion, i.e., a cohesive component/ module focuses on a single function (i.e., single-mindedness) with little interaction with other modules of the system.
In coupling, modules are linked to the other modules.	In cohesion, the module focuses on a single thing.